Exam Traps and Frequently Tested Points: All Topics

Every topic reviewed in this course, in order. Each topic starts with **predicted high-yield points** derived from the past-exam OSCE compilations (2015–2025), followed by the full **Exam Traps and Frequently Tested Points** list from that topic's review file.

How to read the predictions. ★ = this exact point was asked in past compilations, often in more than one year, so expect it again. Likelihood is based on how many years the topic's station appeared: **VERY HIGH** = 3 or more years; **HIGH** = 2 years or embedded in many cases; **MEDIUM** = mostly appears inside other stations.

Prediction Map: Which Topics Are Most Likely

Topic	Likelihood	Evidence from past exams
Acute Kidney Injury	**VERY HIGH**	Contrast AKI station repeated 3 times in 2020; AKI after PCI in cardio 2022–23
Chronic Kidney Disease	**HIGH**	CKD anemia, phosphate and HD dosing appeared in nephrology and critical-care stations
Anemia	**MEDIUM–HIGH**	Anemia on warfarin (nephrology), CAP with anemia on ferrous sulfate (ID), CKD anemia (critical care)
GERD	**HIGH**	GERD full management station 2018–19 and 2020; GERD with latent TB and with asthma
Peptic Ulcer Disease	**VERY HIGH**	H. pylori PUD station repeated 3 times in 2020
Inflammatory Bowel Disease	**VERY HIGH**	Crohn's + TPN 2022–23 and 2025; fulminant UC 2018–19 and 2021; azathioprine 2018–21
Viral Hepatitis	**HIGH**	Chronic HBV with cirrhosis 2020, 2022–23; HCV with cirrhosis 2020
Liver Cirrhosis Complications	**VERY HIGH**	SBP/ascites/HE/HRS stations in 2015, 2018–19, 2020, 2022–23
Pancreatitis	**MEDIUM**	Not a stand-alone past station; appears as a complication inside DM, IBD and nutrition cases
Asthma and COPD	**VERY HIGH**	Asthma and COPD stations in every year 2018–2025; inhaler counseling and smoking cessation
Neurology	**HIGH**	Acute stroke tPA station and carbamazepine–warfarin station 2018–19; meningitis in ID
Psychiatry	**MEDIUM–HIGH**	Depression choice inside COPD, MI and neuropathy cases; insomnia; serotonin syndrome
Rheumatoid Arthritis and Osteoarthritis	**VERY HIGH**	RA on MTX stations in 2018–19, 2020, 2022–23, 2025 (pregnancy planning, monitoring)
Fluids and Electrolytes	**HIGH**	Embedded in DKA, sepsis, HF (K 5.6), TPN and QT cases
Nutrition (EN/PN)	**VERY HIGH**	TPN calculations in Crohn's (2022–23, 2025), HF (2024) and critical care
Thyroid, Osteoporosis and Obesity	**VERY HIGH**	Thyroid storm/myxedema 2018–19, 2020, 2021; hypothyroid dosing yearly; amiodarone thyrotoxicosis 2025
Diabetes	**VERY HIGH**	Insulin stations every year 2018–2023; DKA 2018–2025; new T2DM with HTN 2020
Acute Medical Diseases (PAD, DKA, sepsis, shock, arrest, acid-base)	**VERY HIGH**	Septic shock 2018–19, 2020; DKA; anaphylaxis 2020–21; hemorrhagic shock and acid-base in critical care
DVT and Pulmonary Embolism	**VERY HIGH**	PE in young woman 2022–23 and 2024; pregnancy anti-Xa 2018–19 and 2022–23; cancer VTE; PE with PCOS 2025

Topic	Likelihood	Evidence from past exams
ID 1: UTI, Respiratory Infections, TB, Brucellosis	**VERY HIGH**	TB stations every year; CAP/HAP/VAP 2018–2025; pyelonephritis in pregnancy 2024
ID 2: Endocarditis, SSTI/Osteomyelitis, CNS, IAI, C. difficile	**VERY HIGH**	Meningitis 2018–19 and 2022–23; endocarditis with gentamicin; MRSA osteomyelitis; peritonitis + CDI 2025
ID 3: Fungal Infections and Immunizations	**HIGH**	TB + aspergillosis 2022–23 and 2025; candidemia; vaccines in HTN, COPD and RA stations
Cardio 1: Ischemic Heart Disease	**VERY HIGH**	STEMI/ACS asked in 2015, 2018–19, 2022–23, 2024 and 2025
Cardio 2: Heart Failure, Arrhythmias and AF	**VERY HIGH**	AF then STEMI asked 2018–19, 2021, 2022–23, 2024; HFrEF 2020, 2024, 2025
Cardio 3: Hypertension and Dyslipidemia	**VERY HIGH**	Stage 2 HTN + prediabetes 2020 and 2024; dyslipidemia 2018–19, 2024
Therapeutic Drug Monitoring	**VERY HIGH**	Vancomycin and gentamicin TDM asked in ID and critical care 2018–2025; anti-Xa in pregnancy
Clinical Scores and DOACs	**VERY HIGH**	CHA2DS2-VASc/HAS-BLED in every AF station; Child-Pugh, CURB-65, qSOFA, KDIGO, NYHA in other systems

1. Acute Kidney Injury

Predicted for your exam (likelihood: VERY HIGH). Contrast AKI station repeated 3 times in 2020; AKI after PCI in cardio 2022–23.

1. ★ Contrast-induced AKI prevention: NS **1 mL/kg/h 6–12 h before and after** (0.5 mL/kg/h if EF <35%); urgent case **3 mL/kg/h × 1 h** then 1–1.5 mL/kg/h × 4–6 h.
2. ★ Say clearly: **NAC, sodium bicarbonate and vitamin C have no proven benefit** (PRESERVE); no theophylline, no prophylactic dialysis.
3. ★ Drugs to hold: **metformin (48 h), NSAIDs, ACEi/ARB, MRA, diuretics**, other nephrotoxins (aminoglycosides, vancomycin troughs high).
4. ★ Calculate **FeNa = (UNa × PCr)/(PNa × UCr) × 100**: <1% prerenal, >2% ATN (example 3.4% = ATN, muddy brown casts).
5. ★ KDIGO stage from fold rise (2.1× = stage 2) and urine output; timeline: SCr rises 24–48 h, peaks 3–5 days, recovers 7–14 days.
6. AKI after PCI: keep aspirin, ticagrelor, atorvastatin, β-blocker; renally adjust enoxaparin (q24h if CrCl <30), eptifibatide, rosuvastatin (max 10).
7. Likely add-ons: renal dosing of antibiotics in the same case; when to restart metformin/ACEi (SCr stable); urine output target >0.5 mL/kg/h.

Exam Traps and Frequently Tested Points

1. **FENa < 1% → prerenal**. **> 2% → ATN**.
2. **Patient on diuretics** → use **FEUrea** (< 35% → prerenal).
3. **BUN:SCr > 20 → prerenal**.
4. **Muddy brown casts → ATN**. **WBC casts + eosinophils → AIN**. **RBC casts → GN**.
5. **NSAIDs → afferent vasoconstriction**. **ACEi/ARB → efferent vasodilation**.
6. **NSAID + ACEi/ARB + diuretic → triple whammy**.
7. **Creatinine rise up to 30%** after ACEi → **acceptable**.
8. **Aminoglycosides** → after **5–7 days**, **non-oliguric**, **hypomagnesemia**.
9. **Amphotericin B → saline before and after the dose**, **hypokalemia and hypomagnesemia**.
10. **Cisplatin → hypomagnesemia**.
11. **Vancomycin + pip-tazo → more AKI**.
12. **PPIs and beta-lactams → AIN**.
13. **IV acyclovir → crystals**; prevent with **hydration and slow infusion**.
14. **Trimethoprim, cimetidine, dolutegravir → false rise in creatinine**.

15. **Contrast** → prevent with **saline**; **NAC is not beneficial**.
16. **Metformin + contrast + eGFR < 30** → **hold for 48 hours**.
17. **Loop diuretics** → **only for volume overload**; they **do not treat AKI**.
18. **Low-dose dopamine** → **not beneficial**.
19. **Hydroxyethyl starch (HES)** → **avoid**; it increases AKI.
20. **Hyperkalemia with ECG changes** → **calcium gluconate first**; it **does not lower potassium**.
21. **AEIOU** → dialysis indications.
22. **Loading dose is unchanged** in renal failure.
23. **Cockcroft-Gault is inaccurate** in AKI → assume **CrCl < 10** if creatinine is rising rapidly.
24. **Rasburicase + G6PD deficiency** → **contraindicated**.
25. **Meperidine** → **avoid** in renal failure.

2. Chronic Kidney Disease

Predicted for your exam (likelihood: HIGH). CKD anemia, phosphate and HD dosing appeared in nephrology and critical-care stations.

1. ★ Anemia of CKD: **iron first** (TSAT <20%, ferritin <100 non-HD / <200 HD) → ESA only if Hgb <10; target **10–11.5**, never >13.
2. ★ ESA doses: epoetin 50–100 units/kg 3×/week; darbepoetin 0.45 mcg/kg weekly or 0.75 q2 weeks; weight = dry weight; list causes of hyporesponse.
3. ★ Phosphate ~6 mg/dL: diet 800–1000 mg/day + **sevelamer 800 mg TID with meals**; calcium binders if low Ca.
4. ★ Staging with **CKD-EPI 2021**, drug dosing with **Cockcroft–Gault**; metformin max 1000 at eGFR 30–44, stop <30.
5. HD drug dosing that was asked: vancomycin 25 mg/kg load then 500–1000 mg after HD by pre-HD level; levetiracetam supplement 250–500 mg after HD; pip-tazo 2.25 g q8h + 0.75 g post-HD; apixaban 5 mg BID in HD.
6. Albuminuria A3: max ACEi/ARB + SGLT2i + finerenone; SCr rise up to 30% acceptable.

Exam Traps and Frequently Tested Points

1. **CKD = abnormality for > 3 months** (GFR < 60 **or** markers of damage such as ACR ≥ 30).
2. **Staging uses GFR (G1–G5) and albuminuria (A1–A3).**
3. **Diabetes is the #1 cause**, hypertension #2.
4. **Cardiovascular disease is the #1 cause of death.**
5. **ACEi/ARB** for CKD **with albuminuria**; **creatinine rise up to 30% is acceptable**.
6. **Never combine ACEi + ARB** (or with aliskiren).
7. **SGLT2i** for CKD with **eGFR ≥ 20**, **with or without diabetes**; **continue until dialysis**; initial eGFR dip is expected.
8. **Finerenone:** T2D + albuminuria, **eGFR ≥ 25**, **K ≤ 5.0**; watch **hyperkalemia**.
9. **KDIGO BP target: SBP < 120.**
10. **Metformin:** contraindicated **eGFR < 30**; do not start at **30–44**.
11. **Statin** for CKD **≥ 50 years**; **do not start in dialysis** (but continue).
12. **Protein: 0.8 g/kg/day** (non-dialysis), **1.0–1.2 g/kg/day** (dialysis).
13. **Anemia of CKD** = **↓ EPO**; **correct iron first**.
14. **Iron if TSAT ≤ 30% and ferritin ≤ 500.**
15. **Hemodialysis → IV iron**; non-dialysis → oral iron first.
16. **Iron dextran** → **test dose** (anaphylaxis); **ferumoxytol** → MRI interference; **ferric carboxymaltose** → hypophosphatemia.
17. **Start ESA when Hgb < 10**; **do not exceed 11.5** (reduce if > 11 per FDA).
18. **ESA boxed warning:** stroke, MI, VTE, death, tumor progression; **hypertension** is the most common side effect.
19. **#1 cause of ESA hyporesponsiveness = iron deficiency.**
20. **Do not increase ESA dose more often than every 4 weeks.**

21. **CKD-MBD chain:** ↑ Phos + ↓ vitamin D → ↓ Ca → **↑ PTH**.
22. **All phosphate binders are taken WITH MEALS.**
23. **Calcium carbonate** needs acid (less effective with PPIs); **calcium acetate** is the better binder.
24. **Sevelamer** lowers LDL; **hydrochloride** form causes **acidosis**.
25. **Lanthanum must be chewed.**
26. **Ferric citrate** also treats **iron deficiency**.
27. **Aluminum binders → max 4 weeks**; **never with citrate**.
28. **Elemental calcium from binders ≤ 1,500 mg/day.**
29. **Hypercalcemia** → switch to **non-calcium binder**, reduce VDRA.
30. **Doxercalciferol** needs **liver** activation; **paricalcitol** causes less hypercalcemia than calcitriol.
31. **Cinacalcet/etelcalcetide → dialysis only**; main side effect **hypocalcemia**; **do not combine**; cinacalcet is a **CYP2D6 inhibitor**.
32. **PTH target in dialysis: 2–9 × ULN.**
33. **Treat metabolic acidosis** if bicarbonate **< 18** (some references < 22).
34. **Patiromer** → separate other drugs by **3 hours**.
35. **Loop diuretics** are preferred for edema in advanced CKD; **metolazone** works at low GFR.
36. **Uremic bleeding → desmopressin.**
37. **HD pruritus → difelikefalin** (IV kappa-opioid agonist).
38. **AV fistula** is the preferred HD access; **"save the vein"** in the non-dominant arm.
39. **PD peritonitis → intraperitoneal antibiotics**; **icodextrin → falsely high glucose** readings.
40. **Hepatitis B vaccine → higher dose (40 mcg)** in dialysis patients.
41. **Gadolinium + eGFR < 30 → nephrogenic systemic fibrosis.**
42. **Avoid:** NSAIDs, meperidine, morphine, codeine, glyburide, Fleet enema, magnesium products, nitrofurantoin (< 30), fondaparinux (< 30).
43. **Avoid transfusions** in transplant candidates (**HLA sensitization**).

3. Anemia

Predicted for your exam (likelihood: MEDIUM–HIGH). Anemia on warfarin (nephrology), CAP with anemia on ferrous sulfate (ID), CKD anemia (critical care).

1. ★ Classify by **MCV** and order iron studies; in a warfarin patient look for GI blood loss.
2. ★ Transfusion threshold Hgb **<7** (<8 with cardiac disease).
3. ★ Ferrous sulfate 325 mg (65 mg elemental) **daily or every other day**; separate from doxycycline (2 h before/3–4 h after), quinolones, levothyroxine, PPIs/antacids.
4. Recheck Hgb in 2–4 weeks (expect +1–2 g/dL), continue 3 months after normalization to refill stores.
5. CKD anemia = iron + ESA rules (see CKD); febrile neutropenia G-CSF (filgrastim 5 mcg/kg) is not routine for established FN.

Exam Traps and Frequently Tested Points

1. **Anemia:** Hgb **< 13** (men), **< 12** (women), **< 11** (pregnancy).
2. **Microcytic** (< 80): **TAILS • Macrocytic** (> 100): **B12, folate**, alcohol, drugs.
3. **Ferritin** = best test for iron stores; **low ferritin = iron deficiency**; normal/high in inflammation.
4. **IDA:** ↓ iron, **↑ TIBC**, ↓ TSAT, ↓ ferritin, **↑ RDW • ACD:** ↓ TIBC, normal/↑ ferritin.
5. **Thalassemia:** microcytic, normal iron studies, normal RDW, **Mentzer < 13 → do not give iron**.
6. **B12 deficiency: ↑ MMA + ↑ homocysteine + neurologic symptoms • Folate:** ↑ homocysteine only, normal MMA, no neuro.
7. **Ferrous sulfate 325 mg = 65 mg elemental (20%)** • gluconate 12% • **fumarate 33%**.
8. **Oral iron dose: 60–200 mg elemental/day**; once daily or alternate days is effective with fewer GI effects.
9. Iron: **empty stomach**, **vitamin C ↑**, **avoid antacids, PPIs, calcium, dairy, tea, coffee**.
10. **Iron + tetracyclines/quinolones/levothyroxine/INSTIs** → chelation → **separate**.
11. **Most common iron ADR: constipation**; dark stools are harmless.

12. **Peak reticulocytes 7–10 days**; Hgb ↑ 1–2 g/dL in 2–4 weeks; **continue ~3 months after Hgb normalizes**.
13. **Most common cause of oral iron failure: nonadherence.**
14. **Iron dextran → test dose** (anaphylaxis boxed warning) • **Ferumoxytol → anaphylaxis + MRI interference • Ferric carboxymaltose → hypophosphatemia**.
15. **Iron deficit = wt × (target − actual Hgb) × 2.4 + 500**.
16. **Acute iron poisoning → deferoxamine** (vin rosé urine) • **Deferasirox** → renal/hepatic failure, GI bleed • **Deferiprone → agranulocytosis** (ANC weekly).
17. **Pernicious anemia** = no intrinsic factor → **lifelong B12**; **oral high-dose B12 works** (passive diffusion).
18. **B12 IM: 1,000 mcg daily × 1 week → weekly × 4 → monthly for life.**
19. **Never give folic acid alone in B12 deficiency** (masks it; neuro damage continues).
20. **Hypokalemia** early in B12/folate treatment → monitor K.
21. **Metformin, PPIs, H2RAs** → B12 deficiency.
22. **Methotrexate, trimethoprim, pyrimethamine, phenytoin** → folate deficiency.
23. **Folic acid 1 mg daily × 4 months** for deficiency • **400–800 mcg** for NTD prevention • **4 mg** if previous NTD.
24. **Leucovorin** (not folic acid) for **methotrexate toxicity** and with **pyrimethamine**.
25. **Folic acid ↓ phenytoin levels.**
26. **ESAs:** start Hgb < 10; **avoid Hgb > 11**; boxed warning: death, MI, stroke, VTE, tumor progression; **not when cure is the goal**; **stop after chemo ends**.
27. **Most common ESA ADR: hypertension**; **uncontrolled HTN = contraindication**; most common cause of hyporesponse = **iron deficiency**.
28. **G6PD deficiency → avoid** primaquine, tafenoquine, dapsone, sulfonamides, nitrofurantoin, **rasburicase**, **methylene blue**, fava beans.
29. **Ceftriaxone/cefotetan, methyldopa** → immune hemolytic anemia (Coombs positive).
30. **Warm AIHA → steroids • Cold agglutinin → avoid cold, rituximab** (steroids don't work).
31. **Sickle cell: hydroxyurea** ↑ HbF (myelosuppression, teratogenic, macrocytosis) • **penicillin prophylaxis until age 5 • folic acid** • vaccines • **fever = emergency**.
32. **Sickle cell pain crisis:** fluids + **opioids**; **avoid meperidine • Acute chest syndrome: ceftriaxone + azithromycin** + transfusion.
33. **Voxelotor** was **withdrawn** (2024).
34. **Isoniazid → sideroblastic anemia** → **pyridoxine**.
35. **Chloramphenicol, carbamazepine, felbamate** → aplastic anemia.
36. **1 unit PRBC ↑ Hgb ~1 g/dL**; transfuse at **Hgb < 7** (most patients).
37. **IDA in men/postmenopausal women → rule out GI bleeding/cancer.**

4. GERD

Predicted for your exam (likelihood: HIGH). GERD full management station 2018–19 and 2020; GERD with latent TB and with asthma.

1. ★ Full management: lifestyle (elevate head of bed, avoid late meals, weight loss), then **PPI once daily 30–60 min before breakfast × 8 weeks**, then step down.
2. ★ Drugs that worsen GERD: CCBs, nitrates, anticholinergics, theophylline, benzodiazepines, bisphosphonates, NSAIDs, doxycycline.
3. ★ Long-term PPI risks: C. difficile, fractures, B12 and Mg deficiency, pneumonia; ranitidine withdrawn.
4. ★ Clopidogrel + PPI → **pantoprazole**. Aluminum/magnesium antacids reduce isoniazid absorption (separate ≥1 h).
5. Asthma with GERD: treating GERD only helps if GERD symptoms present; calcium citrate preferred with acid suppression.

Exam Traps and Frequently Tested Points

1. **Most common symptom:** heartburn • **Most common mechanism: transient LES relaxations**.

2. **Alarm symptoms** (dysphagia, odynophagia, weight loss, GI bleeding, anemia) → **endoscopy**, not self-care.
3. **Typical symptoms** → **empiric PPI trial for 8 weeks** (no endoscopy needed first).
4. **Weight loss** has the strongest lifestyle evidence • **Elevate head of bed 6–8 inches** for nighttime symptoms • **No meals 2–3 h before bed**.
5. **Drugs that ↓ LES pressure:** anticholinergics, CCBs, nitrates, theophylline, benzodiazepines, opioids, estrogen/progesterone.
6. **Pill esophagitis:** NSAIDs, bisphosphonates, KCl, iron, doxycycline → full glass of water, stay upright 30 min.
7. **Antacids:** fastest onset, shortest duration, **no healing**.
8. **Aluminum → constipation • Magnesium → diarrhea** • both accumulate in **CKD**.
9. **Sodium bicarbonate** → avoid in HF, HTN, pregnancy • **Calcium carbonate** → milk-alkali syndrome.
10. **Alginic acid (Gaviscon)** → raft, good for **regurgitation**.
11. **Antacids/H2RAs/PPIs + quinolones, tetracyclines, INSTIs, azoles, atazanavir** → ↓ absorption.
12. **H2RAs:** good for **prevention** (30–60 min before meals) and **nocturnal** acid • **tachyphylaxis** in 2 weeks.
13. **Cimetidine** → CYP inhibitor (warfarin, theophylline, phenytoin) + **gynecomastia** → prefer **famotidine**.
14. **H2RAs in elderly/renal impairment** → **confusion** → reduce dose.
15. **Ranitidine withdrawn** (NDMA).
16. **PPIs = most effective**; take **30–60 min before breakfast** (prodrugs need active pumps).
17. **Dexlansoprazole** → can be taken **without regard to meals**; **Zegerid** → empty stomach 1 h before meal.
18. **PPI therapy: once daily × 8 weeks** → partial response → **twice daily** or switch PPI.
19. **OTC PPI: 14 days**, up to **3 courses/year** (every 4 months); ≥ 18 years.
20. **Long-term maintenance PPI: LA C/D esophagitis**, **Barrett's**, strictures.
21. **Long-term PPI ADRs: C. diff**, pneumonia, **fractures**, **hypomagnesemia**, **B12 deficiency**, iron deficiency, AIN.
22. **PPI + calcium:** use **calcium citrate**.
23. **Clopidogrel + omeprazole/esomeprazole** → avoid; use **pantoprazole**.
24. **Rilpivirine + PPI** → **contraindicated**.
25. **Stopping PPIs** → **rebound acid hypersecretion** → **taper**.
26. **No renal dose adjustment** for PPIs; **H2RAs need renal adjustment**.
27. **Vonoprazan** (P-CAB): reversible, not a prodrug, with or without food, 20 mg × 8 weeks for EE.
28. **Metoclopramide:** not for GERD alone; **boxed warning tardive dyskinesia**, max 12 weeks.
29. **Baclofen** → ↓ TLESRs, for refractory regurgitation; CNS side effects.
30. **Sucralfate** → binds drugs (separate 2 h); option in pregnancy.
31. **Pregnancy:** lifestyle → antacids/sucralfate → famotidine → PPI; **avoid sodium bicarbonate and magnesium trisilicate**.
32. **Infants who spit up** ("happy spitters") → **no acid suppression**.
33. **pH monitoring** → stop PPI **7 days** before • **Manometry** before surgery.
34. **Barrett's esophagus** → risk of adenocarcinoma → indefinite PPI + surveillance.
35. **Rule out cardiac causes** of chest pain before attributing it to GERD.

5. Peptic Ulcer Disease

Predicted for your exam (likelihood: VERY HIGH). H. pylori PUD station repeated 3 times in 2020.

1. ★ First-line: **bismuth quadruple × 14 days** (PPI BID + bismuth + tetracycline 500 QID + metronidazole 500 TID–QID).
2. ★ Before UBT/stool antigen: stop **PPI ≥2 weeks**, **antibiotics and bismuth ≥4 weeks**; test of cure **≥4 weeks after** therapy.
3. ★ PPI after therapy: duodenal ulcer 4 weeks, gastric ulcer 8 weeks. Bleeding ulcer: 80 mg IV + 8 mg/h × 72 h.

4. ★ NSAID needed: CV risk → naproxen + PPI; GI risk only → celecoxib + PPI. Ibuprofen blocks aspirin (aspirin 30 min first).
5. Counseling: metronidazole + alcohol, bismuth black stools/tongue, tetracycline with water, adherence for all 14 days.

Exam Traps and Frequently Tested Points

1. **Most common causes: H. pylori** and **NSAIDs**.
2. **Duodenal ulcer:** pain **2–3 h after meals and at night**, **relieved by food**, mostly H. pylori.
3. **Gastric ulcer:** pain **worse with food**, weight loss, **biopsy** (malignancy risk), more NSAIDs.
4. **Healing:** duodenal **4 weeks**, gastric **8 weeks** of PPI.
5. **Most common complication: bleeding**.
6. **NSAID ulcers** are often **painless** until they bleed.
7. **Dyspepsia ≥ 60 → endoscopy**; < 60 → **test and treat** H. pylori.
8. **UBT and stool antigen** = active infection; **serology cannot confirm cure**.
9. **Before testing:** stop **PPIs 2 weeks**, **antibiotics and bismuth 4 weeks**.
10. **Confirm eradication ≥ 4 weeks** after therapy (off PPI 2 weeks).
11. **All H. pylori regimens: 14 days**, **PPI twice daily**.
12. **Bismuth quadruple** (PPI + bismuth + tetracycline + metronidazole) = **preferred first-line** and for **penicillin allergy**.
13. **Clarithromycin triple** only if **clarithromycin susceptibility is known**; avoid after previous macrolide use.
14. **New options: rifabutin triple** (Talicia) and **vonoprazan dual** (vonoprazan + amoxicillin).
15. **Salvage:** don't repeat clarithromycin or levofloxacin • failed BQT → rifabutin or levofloxacin triple.
16. **Bismuth → black tongue/stools** • subsalicylate → avoid in aspirin allergy, warfarin, children (Reye).
17. **Metronidazole → no alcohol** (disulfiram-like) during and **3 days after**; ↑ INR with warfarin; metallic taste.
18. **Tetracycline** → not in pregnancy or < 8 years; photosensitivity; chelation.
19. **Clarithromycin → CYP3A4 inhibitor** (simvastatin, lovastatin, colchicine) + **QT**.
20. **Rifabutin** → orange body fluids, ↓ **oral contraceptives**.
21. **Strongest NSAID ulcer risk factor: previous ulcer/bleed**.
22. **NSAID prevention: PPI preferred**; **misoprostol** effective but diarrhea + **contraindicated in pregnancy**; standard-dose H2RA and sucralfate **do not** prevent NSAID gastric ulcers.
23. **High GI + high CV risk → avoid NSAIDs** • High CV risk needing NSAID → **naproxen + PPI**.
24. **Celecoxib** → less GI risk but ↑ CV risk; GI benefit lost with aspirin.
25. **Clopidogrel + PPI → pantoprazole** (not omeprazole/esomeprazole).
26. **Aspirin for secondary prevention → resume early** after bleeding + PPI.
27. **Ulcer bleeding after endoscopy: PPI 80 mg IV bolus + 8 mg/h × 72 h** (or intermittent high dose), then PO BID to day 14.
28. **Erythromycin IV** before endoscopy improves visualization • **Tranexamic acid** not recommended • **Octreotide** = varices.
29. **Transfuse at Hgb < 7** (< 8 with CVD).
30. **SUP** only for **ICU patients with risk factors** (ventilation > 48 h, coagulopathy) • **stop at discharge**.
31. **ZES** → multiple refractory ulcers + diarrhea + high gastrin → **high-dose PPI**.
32. **Refractory ulcer** → check adherence, H. pylori, hidden NSAIDs, smoking, ZES, cancer.
33. **Smoking** delays ulcer healing • **acetaminophen** is the preferred analgesic.

6. Inflammatory Bowel Disease

Predicted for your exam (likelihood: VERY HIGH). Crohn's + TPN 2022–23 and 2025; fulminant UC 2018–19 and 2021; azathioprine 2018–21.

1. ★ Mild ileal Crohn's: **budesonide 9 mg × 8 weeks**, taper; **mesalamine is not effective in Crohn's**; budesonide is not for maintenance.
2. ★ Moderate–severe or failed azathioprine: **infliximab 5 mg/kg weeks 0, 2, 6 then q8 weeks**; screen TB, HBV, HCV, HIV, varicella; no live vaccines.
3. ★ Azathioprine: TPMT/NUDT15 first; CBC weekly × 4, monthly × 3, then q3 months; pancreatitis; lymphoma/skin cancer; **allopurinol → reduce dose 75%**.
4. ★ Fulminant UC: IV methylprednisolone 60 mg/day (or hydrocortisone 100 q8h), day-3 assessment, rescue infliximab or cyclosporine 2 mg/kg; VTE prophylaxis; avoid loperamide, opioids, anticholinergics, NSAIDs.
5. ★ Crohn's vs UC differences table; perianal fistula metronidazole + ciprofloxacin; rectal mesalamine at bedtime.
6. TPN numbers in the same station (see Nutrition).

Exam Traps and Frequently Tested Points

1. **UC:** colon only, starts at **rectum**, **continuous**, **mucosal**, **bloody diarrhea**, p-ANCA, **colectomy curative**.
2. **CD:** mouth to anus (**terminal ileum**), **skip lesions**, **transmural**, **fistulas/strictures**, granulomas, ASCA.
3. **Smoking protects in UC** but **worsens CD → stop smoking** in CD.
4. **5-ASA** = first-line for **mild–moderate UC** (induction + maintenance); **not for CD**.
5. **Proctitis → mesalamine suppository • Left-sided → enema + oral • Extensive → oral** (± rectal).
6. **Rectal 5-ASA > rectal steroids**; **oral + rectal > oral alone**.
7. **Sulfasalazine:** sulfa allergy, **G6PD hemolysis**, **agranulocytosis**, **oligospermia**, **orange urine/skin**, **folate deficiency → folic acid**.
8. **Mesalamine: interstitial nephritis** → monitor SCr; paradoxical colitis.
9. **Steroids = induction only, never maintenance** • taper; calcium + vitamin D.
10. **Budesonide:** Entocort → **ileocecal CD**; Uceris MMX → **UC**; high first-pass; **avoid CYP3A4 inhibitors/grapefruit**.
11. **Acute severe UC: IV methylprednisolone 60 mg/day** → no response in **3–5 days** → **infliximab or cyclosporine** → colectomy.
12. **Hospitalized IBD → VTE prophylaxis** (even with bleeding) • test for **C. difficile** in flares.
13. **Avoid antidiarrheals, opioids, anticholinergics** in severe colitis (**toxic megacolon**) • **avoid NSAIDs** (use acetaminophen).
14. **Thiopurines:** slow (3–6 months) → **maintenance • check TPMT/NUDT15** • myelosuppression, pancreatitis, lymphoma, skin cancer.
15. **Allopurinol/febuxostat + azathioprine/6-MP** → severe myelosuppression → **reduce to 25–33%**.
16. **Methotrexate: CD only**, **once weekly**, folic acid, **teratogenic** (stop 3 months before conception), avoid TMP-SMX.
17. **Cyclosporine:** rescue in acute severe UC • nephrotoxicity, HTN, hypomagnesemia, seizures • monitor levels.
18. **Anti-TNF: screen TB and HBV • contraindicated in NYHA III–IV HF** and **demyelinating disease** • drug-induced lupus, psoriasis.
19. **Certolizumab = CD only**, minimal placental transfer • **Golimumab = UC only** • **Infliximab** = best for **fistulas**.
20. **Infliximab + thiopurine** (combination) → more effective, ↓ antibodies; watch **HSTCL** in young men.
21. **Vedolizumab: gut-selective**, safest, slower onset • **Natalizumab:** PML (JC virus).
22. **Ustekinumab:** IL-12/23, **single weight-based IV dose** then SC every 8 weeks • **Risankizumab:** monitor LFTs.

23. **JAK inhibitors** (tofacitinib = UC only; upadacitinib = UC + CD): **boxed: infection, death, cancer, MACE, thrombosis • after ≥ 1 TNF failure • herpes zoster** (Shingrix) • lipids.
24. **Tofacitinib + strong CYP3A4 inhibitor** → reduce dose; avoid with strong inducers.
25. **S1P modulators** (ozanimod, etrasimod; UC only): **first-dose bradycardia**, **macular edema**, LFTs • ozanimod + **MAOIs** contraindicated.
26. **Perianal CD** → metronidazole/ciprofloxacin + drainage + infliximab • **Pouchitis** → cipro/metronidazole.
27. **No live vaccines** on immunosuppressants (prednisone ≥ 20 mg ≥ 2 weeks) • give inactivated vaccines + Shingrix.
28. **Colonoscopy surveillance 8 years** after diagnosis; yearly with **PSC**.
29. **Pregnancy:** continue 5-ASA, thiopurines, anti-TNF, vedolizumab, ustekinumab • **MTX contraindicated** • avoid JAKi and S1P.
30. **Ileal CD/resection** → **B12** deficiency, **bile-acid diarrhea** (cholestyramine), **oxalate kidney stones**, gallstones.

7. Viral Hepatitis

Predicted for your exam (likelihood: HIGH). Chronic HBV with cirrhosis 2020, 2022–23; HCV with cirrhosis 2020.

1. ★ HBV: **entecavir 0.5 mg (1 mg decompensated) on an empty stomach**, renally adjusted; TDF 300; TAF 25 (not in Child B/C). **Never stop abruptly** (flare).
2. ★ TDF monitoring: SCr, phosphate, urine glucose/protein, bone density. Peginterferon contraindicated in decompensation and severe depression.
3. ★ HBV pregnancy: TDF from week 28 if HBV DNA >200,000; infant HBIG + vaccine at birth.
4. ★ Decompensated HCV: **sofosbuvir/velpatasvir + ribavirin × 12 weeks** (or 24 weeks without RBV); no protease inhibitors in Child B/C; SVR12 = cure.
5. Velpatasvir + PPI: with food, 4 h before omeprazole 20 mg (or H2RA ≤ famotidine 40 BID). Screen HBV before DAAs and before biologics (reactivation).

Exam Traps and Frequently Tested Points

1. **A and E** = **fecal-oral**, acute only • **B, C, D** = blood-borne, chronic.
2. **HBV chronicity**: **90% newborns** vs **< 5% adults** • **HCV chronicity**: **75–85%**.
3. **HEV in pregnancy** → fulminant hepatitis, high mortality.
4. **HDV** needs **HBsAg**; superinfection is worse; **HBV vaccine prevents HDV**.
5. **HCV has no vaccine** and no PEP • **HAV and HBV have vaccines**.
6. **IgM anti-HAV** = acute HAV • **IgM anti-HBc** = acute HBV.
7. **HBsAg > 6 months** = chronic HBV • **anti-HBs alone** = vaccinated • **anti-HBc + anti-HBs** = recovered.
8. **Anti-HBc is NOT produced by vaccination** (only infection).
9. **HBeAg** = high replication and infectivity.
10. **HAV vaccine:** 2 doses (0, 6–12/18 months) • PEP within **2 weeks**: vaccine (healthy ≥ 1 year), **IG 0.1 mL/kg** (< 12 months), both if immunocompromised/chronic liver disease.
11. **HBV vaccine:** Engerix-B/Recombivax 0, 1, 6 months • **Heplisav-B 2 doses (0, 1 month)** • **dialysis 40 mcg** • **birth dose within 24 h** • protective anti-HBs **≥ 10 mIU/mL**.
12. **Infant of HBsAg+ mother: HBIG + vaccine within 12 hours** • test at 9–12 months • **breastfeeding OK**.
13. **Mother HBV DNA > 200,000 → TDF from 28–32 weeks**.
14. **HBV first-line drugs: entecavir**, **TDF**, **TAF** (and peginterferon) • **avoid lamivudine** (resistance) and adefovir.
15. **Entecavir: empty stomach** • 0.5 mg (naïve), **1 mg** (lamivudine-resistant, decompensated) • avoid if lamivudine resistance.
16. **TDF: kidney (Fanconi) and bone** toxicity • **preferred in pregnancy** • **TAF**: less kidney/bone toxicity, take with food.

17. **Stopping HBV antivirals → severe hepatitis flare** (boxed) • **test HIV** before starting (resistance).
18. **Peginterferon:** finite, no resistance • **contraindicated in decompensated cirrhosis, pregnancy, severe depression** • flu-like symptoms, depression, thyroid problems.
19. **Screen HBsAg + anti-HBc before** chemotherapy, **rituximab**, anti-TNF, and **HCV DAAs** → prophylaxis with entecavir/tenofovir.
20. **HCV screening:** all adults once + every pregnancy • **antibody → confirm with HCV RNA**.
21. **HCV antibody stays positive after cure** → use RNA for reinfection.
22. **Cure = SVR12** (undetectable HCV RNA 12 weeks after treatment).
23. **Mavyret: 3 tablets daily with food × 8 weeks • Epclusa: 1 tablet daily × 12 weeks** — both pangenotypic.
24. **Protease inhibitors (-previr) contraindicated in decompensated cirrhosis** → use **sofosbuvir/velpatasvir + ribavirin**.
25. **Sofosbuvir + amiodarone** → severe bradycardia.
26. **Velpatasvir/ledipasvir + PPIs/H2RAs/antacids** → ↓ absorption (special timing rules).
27. **Rifampin, carbamazepine, phenytoin, St. John's wort + DAAs** → contraindicated.
28. **Mavyret + ethinyl estradiol** → ALT elevation; **Mavyret + atorvastatin/simvastatin** → avoid.
29. **Ribavirin:** hemolytic anemia + **teratogenic** (2 contraceptives, 6 months after, both partners) • never monotherapy.
30. **After HCV cure:** reinfection possible; cirrhosis still needs **HCC surveillance every 6 months**.
31. **Chronic liver disease:** acetaminophen ≤ 2 g/day, avoid NSAIDs and alcohol, vaccinate HAV/HBV.
32. **ALT > AST** in viral hepatitis; **AST:ALT > 2** in alcoholic hepatitis.

8. Liver Cirrhosis Complications

Predicted for your exam (likelihood: VERY HIGH). SBP/ascites/HE/HRS stations in 2015, 2018–19, 2020, 2022–23.

1. ★ SBP: neutrophils ≥250; **ceftriaxone 2 g × 5–7 days + albumin 1.5 g/kg day 1 and 1 g/kg day 3**; secondary prophylaxis ciprofloxacin 500 daily.
2. ★ HRS: stop diuretics/NSBB/NSAIDs; albumin 1 g/kg then 20–40 g/day + **terlipressin 0.85 mg q6h**; SpO2 must be ≥90%; day-4 response rules.
3. ★ Ascites: Na 2 g, **spironolactone 100 + furosemide 40** (max 400/160); urine Na interpretation; LVP >5 L → albumin 6–8 g/L.
4. ★ HE: lactulose 25 mL q1–2h → 2–3 soft stools/day; rifaximin 550 BID for recurrence; **no protein restriction**.
5. ★ Varices: octreotide 50 mcg + 50 mcg/h, ceftriaxone 1 g, EBL ≤12 h, Hgb target 7–8; secondary NSBB + EBL; hold carvedilol if SBP <90/AKI.
6. Child-Pugh calculation; dose adjustments in cirrhosis (escitalopram max 10, avoid PZA, metronidazole reduce).

Exam Traps and Frequently Tested Points

1. **Child-Pugh:** bilirubin, albumin, INR, ascites, encephalopathy (A 5–6, B 7–9, C 10–15) • **MELD:** bilirubin, INR, creatinine, Na → transplant priority.
2. **Ascites:** sodium **2 g/day**; **fluid restriction only if Na < 125**.
3. **Diuretics: spironolactone 100 mg + furosemide 40 mg** (100:40 ratio), max **400:160**, titrate every 3–5 days.
4. **Weight loss goal:** 0.5 kg/day (no edema), 1 kg/day (edema).
5. **Gynecomastia** from spironolactone → **amiloride** or eplerenone.
6. **Tense ascites → LVP • > 5 L removed → albumin 6–8 g/L**.
7. **SAAG ≥ 1.1** = portal hypertension.
8. **Avoid in ascites:** NSAIDs, ACEi/ARBs, aminoglycosides • stop NSBB if SBP < 90, Na < 130, or AKI.
9. **Tolvaptan** → boxed warning: avoid in liver disease • correct Na ≤ 8 mEq/L/24 h.
10. **SBP diagnosis: ascitic PMN ≥ 250 cells/mm³** • most common organism **E. coli**.

11. **SBP treatment: cefotaxime 2 g q8h** or **ceftriaxone** × 5 days + **albumin 1.5 g/kg day 1 and 1 g/kg day 3** (if SCr > 1, BUN > 30, or bilirubin > 4).
12. **Albumin in SBP ↓ HRS and mortality.**
13. **GI bleed in cirrhosis → ceftriaxone 1 g daily ≤ 7 days.**
14. **After SBP → lifelong ciprofloxacin or TMP-SMX daily** (secondary prophylaxis).
15. **Primary SBP prophylaxis:** ascitic protein < 1.5 + renal dysfunction or liver failure.
16. **NSBBs** (propranolol, nadolol, carvedilol) → HR **55–60**, SBP ≥ 90 • **selective β1-blockers don't work** • **no nitrates**.
17. **Carvedilol** → greatest portal pressure ↓ (α1 + β), max 12.5 mg/day.
18. **Primary prophylaxis (medium/large varices):** NSBB **or** EVL (not both).
19. **Secondary prophylaxis:** NSBB **+** EVL (both).
20. **Acute variceal bleed: octreotide** (50 mcg bolus + 50 mcg/h × 2–5 days) + **ceftriaxone** + **EVL within 12 h** + **restrictive transfusion (Hgb 7)** • **hold NSBB** during the bleed.
21. **Pre-emptive TIPS** within 72 h for Child C 10–13 or Child B with active bleeding.
22. **TIPS** ↑ hepatic encephalopathy risk.
23. **HE: find the precipitant first** (infection, GI bleed, constipation, dehydration, hypokalemia, benzodiazepines/opioids).
24. **Ammonia levels not needed** for diagnosis or monitoring.
25. **Lactulose** first-line → titrate to **2–3 soft stools/day** • too much → dehydration, hypernatremia → worse HE.
26. **Rifaximin 550 mg BID** → add to lactulose for **recurrent** HE.
27. **Do NOT restrict protein in HE** (1.2–1.5 g/kg/day).
28. **Hypokalemia and alkalosis worsen HE** (↑ ammonia).
29. **HRS:** functional, kidneys normal, **no response to albumin 1 g/kg × 2 days**, urine Na < 10, no nephrotoxins, no shock.
30. **HRS treatment: terlipressin + albumin** (boxed: **respiratory failure**; avoid if SpO2 < 90% or volume overload) • norepinephrine + albumin (ICU) • midodrine + octreotide + albumin • **liver transplant is definitive**.
31. **Most common HRS precipitant: SBP.**
32. **Acetaminophen ≤ 2 g/day** is the safest analgesic; **avoid NSAIDs**.
33. **INR does not predict bleeding** in cirrhosis; give VTE prophylaxis when hospitalized.

9. Pancreatitis

Predicted for your exam (likelihood: MEDIUM). Not a stand-alone past station; appears as a complication inside DM, IBD and nutrition cases.

1. Drug causes likely asked: **azathioprine**, GLP-1 RA/DPP-4i, valproate, thiazides, didanosine, estrogens; hypertriglyceridemia >1000.
2. Prior pancreatitis → avoid GLP-1 RA and DPP-4i in diabetes.
3. PN: hold lipid if TG >400 mg/dL; TG ≥500 → fibrate to prevent pancreatitis.
4. Acute management: lactated Ringer's (moderate, goal-directed), early oral/enteral feeding, no prophylactic antibiotics, opioids acceptable for pain.

Exam Traps and Frequently Tested Points

1. **Acute pancreatitis diagnosis: 2 of 3** — typical pain, **lipase/amylase ≥ 3 × ULN**, imaging.
2. **Lipase** is more specific than amylase; enzyme levels **do not reflect severity**.
3. **Most common causes: gallstones** > alcohol > hypertriglyceridemia (**TG > 1,000**).
4. **Ultrasound** in all patients (gallstones) • CT only if unclear or no improvement in 48–72 h.
5. **Drug causes: azathioprine/6-MP**, **valproic acid**, **didanosine**, **pentamidine**, **TMP-SMX**, **tetracyclines**, **furosemide/thiazides**, **mesalamine/sulfasalazine**, **estrogens**, **L-asparaginase**, **GLP-1 RAs/DPP-4 inhibitors**, ACE inhibitors, propofol.
6. **Severity:** mild (no organ failure), moderately severe (organ failure < 48 h), **severe (persistent organ failure > 48 h)**.

7. **BISAP ≥ 3**, persistent SIRS, rising BUN, Hct > 44% → severe disease.
8. **Fluids: Lactated Ringer's preferred • moderate** (10 mL/kg bolus if hypovolemic + 1.5 mL/kg/h), not aggressive.
9. **Pain:** opioids (hydromorphone, morphine, fentanyl) are fine • **avoid meperidine**.
10. **Mild AP: early oral feeding (24–48 h)**, low-fat solid diet — don't wait for lipase.
11. **Severe AP: enteral > parenteral nutrition** (NG or NJ).
12. **No prophylactic antibiotics** (even severe or sterile necrosis) • treat only **infected necrosis** or other infections → **carbapenem** or **quinolone + metronidazole**.
13. **Infected necrosis:** antibiotics + **delayed (> 4 weeks) minimally invasive drainage**.
14. **Gallstone AP: ERCP within 24 h only with cholangitis • cholecystectomy same admission** (mild).
15. **Hypertriglyceridemia AP: insulin infusion** (+ dextrose), plasmapheresis • then fibrates, omega-3, low-fat diet.
16. **Post-ERCP pancreatitis prevention: rectal indomethacin 100 mg** + LR hydration + pancreatic stent.
17. **Not recommended:** octreotide, probiotics, prophylactic antibiotics, prolonged NPO.
18. **Hypocalcemia** in AP from fat saponification; Cullen and Grey Turner signs = hemorrhagic pancreatitis.
19. **Pseudocyst** = > 4 weeks, fluid only • asymptomatic → observe.
20. **Chronic pancreatitis: alcohol** (#1) + **smoking** • triad: **calcifications, steatorrhea, diabetes**.
21. **Fecal elastase < 200 mcg/g** → exocrine insufficiency • amylase/lipase often normal in CP.
22. **Type 3c diabetes** → loss of glucagon → **hypoglycemia risk** • metformin first, insulin often needed.
23. **CP pain:** acetaminophen/NSAIDs → **pregabalin** (best evidence), TCAs, SNRIs → **tramadol** → strong opioids last.
24. **PERT (pancrelipase)** dosed by **lipase units**: adults **40,000–50,000 units/meal**, **half with snacks**.
25. **Max 2,500 lipase units/kg/meal** (10,000 units/kg/day) → **fibrosing colonopathy** at high doses.
26. **Take PERT with the first bite** of each meal/snack • swallow whole • can sprinkle on **applesauce** (not dairy) • don't hold in mouth.
27. **Poor PERT response** → check adherence/timing, ↑ dose, **add PPI**, rule out SIBO.
28. **Viokace** (non-enteric-coated) **requires a PPI** • PERT ↓ effect of **acarbose/miglitol**.
29. **PERT is porcine** (religious considerations).
30. **Supplement vitamins A, D, E, K**; check **bone density**.
31. **Autoimmune (IgG4) pancreatitis → corticosteroids**.

10. Asthma and COPD

Predicted for your exam (likelihood: VERY HIGH). Asthma and COPD stations in every year 2018–2025; inhaler counseling and smoking cessation.

1. ★ Asthma: never SABA-only; **budesonide-formoterol MART** (step 3: 1 inhalation BID; step 4: 2 BID + PRN; max 12/day). Check technique/adherence first.
2. ★ COPD: GOLD grade (FEV1) + group A/B/E → treatment; ICS only if eosinophils ≥300 (or ≥100 with exacerbations); dupilumab/mepolizumab for eos-high.
3. ★ COPD exacerbation: O2 88–92%, NIV (pH ≤7.35, PaCO2 ≥45), **prednisone 40 mg × 5 days**, antibiotics by cardinal symptoms × 5–7 days.
4. ★ Device counseling: MDI (prime, spacer, 1 min between puffs), Diskus (no spacer, rinse mouth), HandiHaler, Respimat.
5. ★ Smoking cessation: varenicline titration, bupropion SR (seizures, eating disorders), NRT patch 21/14/7 and gum 2 vs 4 mg.
6. Vaccines: influenza, PCV20/21, RSV ≥60–75, Tdap, zoster. COPD + depression with weight loss → mirtazapine.

Exam Traps and Frequently Tested Points

1. **Asthma** = variable, reversible (FEV1 ↑ > 12% and 200 mL) • **COPD** = post-bronchodilator **FEV1/FVC < 0.70**.

2. **SABA-only** treatment in asthma is **no longer recommended** — all patients need ICS.
3. **GINA preferred reliever: as-needed low-dose ICS-formoterol** (steps 1–2) and **MART** (steps 3–5).
4. **MART only with formoterol** (fast onset) — **never with salmeterol**; max **12 inhalations/day**.
5. **LABA monotherapy in asthma is contraindicated** (↑ deaths) • allowed in COPD.
6. **Uncontrolled:** > 2 days/week symptoms or reliever use, any night waking, activity limitation • **check technique and adherence before stepping up** • step down after 3 months of control.
7. **Asthma exacerbation:** albuterol every 20 min × 3, **ipratropium in ED (moderate–severe)**, **prednisone 40–50 mg × 5–7 days** (no taper), **IV magnesium 2 g** if severe • O2 93–95% • **no antibiotics/theophylline**.
8. **Peak flow:** green ≥ 80%, yellow 50–79%, red < 50%.
9. **EIB:** SABA (or ICS-formoterol) **5–20 minutes before exercise**.
10. **AERD:** avoid aspirin/NSAIDs; montelukast helps.
11. **Pregnancy:** continue therapy; **budesonide** preferred ICS.
12. **Biologics:** omalizumab (anti-IgE, allergic, **anaphylaxis**) • mepolizumab, reslizumab (IV, anaphylaxis), benralizumab (anti-IL-5/IL-5R, eosinophilic) • dupilumab (IL-4Rα; conjunctivitis) • **tezepelumab (TSLP; any phenotype)**.
13. **Montelukast:** boxed warning — **neuropsychiatric events** (suicidal thoughts).
14. **Zafirlukast/zileuton** → hepatotoxicity • ↑ warfarin (and zileuton ↑ theophylline).
15. **Theophylline 5–15 mcg/mL** • CYP1A2: **cipro, fluvoxamine**, macrolides, cimetidine, **quitting smoking** ↑ levels.
16. **COPD groups: A** → a bronchodilator • **B** → **LABA + LAMA** • **E** → **LABA + LAMA** (+ ICS if eos ≥ 300).
17. **ICS-LABA not recommended in COPD** → use triple if ICS indicated.
18. **ICS in COPD:** eos **≥ 300** strongly favors; **< 100** or **repeated pneumonia** → avoid/withdraw.
19. **Roflumilast:** FEV1 < 50% + chronic bronchitis; **diarrhea, weight loss, psychiatric effects**; avoid strong CYP inducers; not a bronchodilator.
20. **Azithromycin** long term → QT, hearing loss • best in former smokers.
21. **Dupilumab** and **mepolizumab** now approved for **eosinophilic COPD** • **ensifentrine** = nebulized PDE3/4 inhibitor.
22. **Smoking cessation** = most important COPD intervention • **varenicline** most effective • **bupropion** contraindicated in seizures/eating disorders.
23. **LTOT:** PaO2 ≤ 55 or SaO2 ≤ 88%, ≥ 15 h/day → ↑ survival.
24. **COPD exacerbation:** SABA ± SAMA, **prednisone 40 mg × 5 days**, antibiotics if purulent sputum (5–7 days), **O2 88–92%**, **NIV** for respiratory acidosis.
25. **Vaccines:** influenza, pneumococcal, COVID-19, **RSV (≥ 60)**, Tdap, zoster.
26. **ICS ADRs:** thrush, dysphonia → **rinse and spit**, use spacer • COPD → ↑ **pneumonia**.
27. **Ritonavir/cobicistat + fluticasone/budesonide** → Cushing's → use **beclomethasone**.
28. **Beta-agonist ADRs:** tremor, tachycardia, **hypokalemia**, hyperglycemia.
29. **Anticholinergic ADRs:** dry mouth, urinary retention (BPH), glaucoma.
30. **Non-selective beta-blockers** (including timolol eye drops) → avoid in asthma; cardioselective OK in COPD.
31. **MDI:** shake, inhale **slowly**, hold 10 s, spacer • **DPI:** no shake, inhale **fast and forceful**, no spacer, don't exhale into device.
32. **Asthma-COPD overlap** → always include ICS.

11. Neurology

Predicted for your exam (likelihood: HIGH). Acute stroke tPA station and carbamazepine–warfarin station 2018–19; meningitis in ID.

1. ★ tPA exclusions and dose (0.9 mg/kg, max 90, 10% bolus); BP <185/110 before, <180/105 after; no antithrombotics × 24 h after tPA.
2. ★ Minor stroke/high-risk TIA: clopidogrel 300 → 75 + aspirin × 21 days; atorvastatin 80; AF stroke → DOAC timing (48 h minor, day 6–7 major).

3. ★ Carbamazepine induces CYP3A4/2C9 → INR falls → prefer **levetiracetam 500 mg BID**; if kept, raise warfarin and check INR weekly.
4. Status epilepticus: lorazepam 4 mg IV (or midazolam 10 mg IM) → levetiracetam 60 mg/kg / fosphenytoin 20 mg PE/kg / valproate 40 mg/kg.
5. Carbapenems drop valproate levels → breakthrough seizures. Levetiracetam: renal dose and post-HD supplement.
6. Neuropathic pain: duloxetine 60 / pregabalin / gabapentin (renal dosing).

Exam Traps and Frequently Tested Points

Epilepsy

1. **Absence → ethosuximide** (or valproate) • **avoid carbamazepine, oxcarbazepine, phenytoin, gabapentin** in absence/myoclonic.
2. **Valproate**: most effective for generalized seizures but **avoid in women of childbearing potential** (neural tube defects, ↓ IQ).
3. **Lamotrigine**: slow titration (SJS) • **halve dose with valproate • estrogen ↓ levels**.
4. **Carbamazepine**: HLA-B*1502 → SJS • autoinduction • SIADH • aplastic anemia.
5. **Phenytoin**: non-linear kinetics, **albumin correction**, gingival hyperplasia • IV max 50 mg/min.
6. **Levetiracetam**: behavior changes; renal dosing; few DDIs.
7. **Topiramate**: kidney stones, metabolic acidosis, glaucoma, cognitive slowing.
8. **Carbapenems ↓ valproate**.
9. **Enzyme inducers ↓ oral contraceptives** → IUD/depot.
10. **Status epilepticus: benzodiazepine first** (lorazepam IV, midazolam IM) → fosphenytoin/valproate/levetiracetam → anesthetic infusion.
11. Isoniazid seizures → **pyridoxine**.

Stroke

1. **Non-contrast CT first** to rule out hemorrhage.
2. **Alteplase 0.9 mg/kg (max 90 mg), 10% bolus** • tenecteplase 0.25 mg/kg bolus (max 25 mg) • within **3–4.5 h**.
3. **BP < 185/110 before**, **< 180/105 for 24 h after • no antithrombotics for 24 h**.
4. **No tPA:** permissive HTN up to 220/120 • **aspirin within 24–48 h**.
5. **Minor stroke/high-risk TIA → aspirin + clopidogrel × 21 days**.
6. **AF → DOAC** (not aspirin) • **high-intensity statin** • BP < 130/80.
7. **SAH → nimodipine 60 mg q4h × 21 days** (never IV).

Parkinson's disease

1. **TRAP • carbidopa/levodopa** most effective • carbidopa prevents peripheral conversion.
2. **Dopamine agonists** → sleep attacks, **impulse-control disorders**, hallucinations, edema.
3. **Selegiline** → insomnia • **MAO-B inhibitors + meperidine** → contraindicated.
4. **Entacapone** → orange urine, diarrhea • **tolcapone** → hepatotoxicity.
5. **Amantadine** → dyskinesia, livedo reticularis.
6. **Anticholinergics** → tremor in young only; avoid in elderly.
7. **Apomorphine + ondansetron** → contraindicated (hypotension).
8. **PD psychosis → pimavanserin**, quetiapine, clozapine • avoid haloperidol, risperidone, metoclopramide.
9. **Rivastigmine** → PD dementia.

Headache

1. **Triptans CI:** CAD, stroke, uncontrolled HTN, hemiplegic migraine, MAOIs, ergots within 24 h.
2. **Rizatriptan 5 mg with propranolol • eletriptan + strong CYP3A4 inhibitors** CI.
3. **Gepants and lasmiditan** → safe with CV disease • lasmiditan: no driving 8 h.
4. **MOH:** limit acute therapy to ≤ 2–3 days/week; triptans/opioids ≥ 10 days/month.
5. **Prevention** if ≥ 4 headache days/month • propranolol, topiramate, CGRP mAbs, gepants • botulinum toxin for chronic migraine.

6. **Erenumab** → constipation, hypertension.
7. **Migraine with aura → no estrogen contraceptives**.
8. **Cluster:** 100% O2 + SC sumatriptan • **verapamil** prevention (ECG).

Multiple sclerosis

1. **Relapse → methylprednisolone 1 g IV × 3–5 days**.
2. **Interferons** → flu-like symptoms, depression • **glatiramer** → post-injection reaction, lipoatrophy.
3. **Teriflunomide** → hepatotoxicity, teratogenic → cholestyramine washout.
4. **Dimethyl fumarate** → flushing (aspirin), lymphopenia.
5. **Fingolimod** → first-dose bradycardia (6-h monitoring), macular edema.
6. **Natalizumab** → **PML** (JCV antibodies).
7. **Ocrelizumab** → only DMT for **PPMS** • HBV screen.
8. **Alemtuzumab** → autoimmunity (thyroid, ITP).
9. **Dalfampridine** → seizures; CI CrCl ≤ 50.
10. **Tizanidine + ciprofloxacin/fluvoxamine** → contraindicated • **baclofen** → don't stop abruptly.

Dementia

1. **Rule out reversible causes** (B12, thyroid, depression, drugs).
2. **Cholinesterase inhibitors** → mild–severe AD • GI effects, **bradycardia/syncope**.
3. **Donepezil at bedtime** (morning if vivid dreams) • **rivastigmine with food**; patch fewer GI effects; re-titrate if missed > 3 days.
4. **Memantine** → moderate–severe; renal adjustment (CrCl < 30).
5. **Anticholinergics cancel cholinesterase inhibitors** → deprescribe (diphenhydramine, oxybutynin).
6. **Lecanemab/donanemab** → early AD with amyloid; **ARIA** (MRI, ApoE ε4).
7. **Antipsychotics in dementia** → boxed ↑ mortality; **brexpiprazole** approved for AD agitation.
8. **Lewy body dementia** → avoid typical antipsychotics.

12. Psychiatry

Predicted for your exam (likelihood: MEDIUM–HIGH). Depression choice inside COPD, MI and neuropathy cases; insomnia; serotonin syndrome.

1. ★ After MI or in HF: **sertraline** (SADHART); avoid TCAs, high-dose citalopram (QT), venlafaxine (BP). Taper SNRIs (75 → 37.5 mg × 1–2 weeks).
2. ★ Serotonin syndrome: SSRI/SNRI + linezolid, tramadol, triptans; MAOI washout 2 weeks (fluoxetine 5 weeks).
3. ★ Mirtazapine for depression with insomnia and weight loss; escitalopram max 10 mg in elderly/hepatic impairment.
4. SSRI + antiplatelet/anticoagulant → bleeding → add PPI; fluoxetine inhibits CYP2C9 (INR up) and 2D6.
5. Insomnia: CBT-I first; zolpidem 5 mg (women/elderly); avoid benzodiazepines in elderly (Beers).
6. Duloxetine: SIADH/hyponatremia, BP, LFTs; avoid CrCl <30.

Exam Traps and Frequently Tested Points

Schizophrenia

1. **≥ 2 symptoms for 1 month** (one = delusions, hallucinations, or disorganized speech) + **6 months** total.
2. **High-potency FGAs** → EPS • **low-potency** → sedation, anticholinergic, orthostasis.
3. **Risperidone/paliperidone** → prolactin • **clozapine/olanzapine** → metabolic • **aripiprazole** → akathisia, ↓ prolactin.
4. **Ziprasidone** (≥ 500 kcal) and **lurasidone** (≥ 350 kcal) → take with food • ziprasidone → QT.
5. **Clozapine** → treatment-resistant (≥ 2 failed trials) • neutropenia (ANC), seizures, myocarditis, constipation • **quitting smoking ↑ levels**.

6. **Dystonia → diphenhydramine/benztropine IM • akathisia → propranolol • TD → valbenazine/deutetrabenazine** (anticholinergics worsen TD).
7. **NMS:** fever, lead-pipe rigidity, ↑ CK → stop drug, dantrolene/bromocriptine.
8. **Boxed:** ↑ mortality in elderly with dementia.
9. **Metabolic monitoring** at baseline, 12 weeks, then yearly.

MDD

1. **SIG E CAPS**, ≥ 5 symptoms for ≥ 2 weeks • **screen for bipolar** before antidepressants.
2. **Trial 4–8 weeks** • continue **4–9 months after remission** • ≥ 3 episodes → long term.
3. **Boxed suicidality < 25 years**.
4. **Citalopram max 40 mg (20 mg if > 60)** — QT.
5. **Paroxetine** → anticholinergic, worst discontinuation, avoid pregnancy, ↓ tamoxifen.
6. **Fluoxetine** → long half-life, 5-week MAOI washout.
7. **Fluvoxamine + tizanidine** → contraindicated (CYP1A2).
8. **Bupropion** → no sexual dysfunction/weight gain • CI seizures, eating disorders.
9. **Mirtazapine** → sedation, weight gain • **trazodone** → priapism.
10. **Venlafaxine** → ↑ BP • **duloxetine** → hepatotoxicity.
11. **TCAs** → lethal overdose (sodium bicarbonate), avoid elderly.
12. **MAOIs** → tyramine, sympathomimetics, serotonergic drugs (linezolid, meperidine, dextromethorphan) • 2-week washout.
13. **Serotonin syndrome** (rapid, clonus, hyperreflexia) vs **NMS** (slow, rigidity) • cyproheptadine.
14. **Esketamine** → REMS, 2-hour observation • **zuranolone** → postpartum depression, no driving 12 h.
15. **Pregnancy → sertraline • children → fluoxetine (≥ 8), escitalopram (≥ 12)**.

Bipolar disorder

1. **Mania ≥ 1 week**, **hypomania ≥ 4 days** • DIGFAST • bipolar I = mania; bipolar II = hypomania + depression.
2. **Avoid antidepressant monotherapy**.
3. **Lithium levels:** acute 0.8–1.2, maintenance 0.6–1.0 • trough 12 h, 5 days after change.
4. **Lithium ↑ by thiazides, ACEi/ARBs, NSAIDs, dehydration** • ↓ by caffeine, theophylline.
5. **Lithium ADRs:** tremor, nephrogenic DI, hypothyroidism, Ebstein's anomaly, CKD.
6. **Valproate** → mixed/rapid cycling • avoid in women of childbearing potential.
7. **Lamotrigine** → maintenance (prevents depression), not acute mania.
8. **Bipolar depression:** quetiapine, lurasidone, cariprazine, lumateperone, olanzapine/fluoxetine.

Anxiety

1. **SSRIs/SNRIs + CBT** first-line • start low in panic • 8–12 weeks for full effect.
2. **Buspirone** → GAD only, delayed onset, not PRN, no dependence.
3. **Benzodiazepines** → short-term bridge • boxed: opioids, abuse, dependence • taper slowly.
4. **LOT** (lorazepam, oxazepam, temazepam) → liver disease and elderly.
5. **Performance anxiety → propranolol • PTSD nightmares → prazosin • avoid benzodiazepines in PTSD**.
6. **OCD** → higher SSRI doses, 8–12 weeks, clomipramine.

Insomnia

1. **CBT-I** first-line.
2. **Z-drugs** → complex sleep behaviors (boxed) • **zolpidem 5 mg in women/elderly** • eszopiclone metallic taste.
3. **DORAs** (suvorexant) → maintenance; CI narcolepsy.
4. **Ramelteon** → onset, no abuse • CI fluvoxamine.
5. **Low-dose doxepin** → maintenance, OK in elderly.
6. **Avoid diphenhydramine and benzodiazepines in elderly**.
7. **Trazodone, melatonin, OTC antihistamines** → not recommended by AASM for chronic insomnia.

Substance use disorders

1. **≥ 2 of 11 criteria** • 2–3 mild, 4–5 moderate, ≥ 6 severe • tolerance/withdrawal don't count if prescribed.
2. **Alcohol withdrawal → benzodiazepines** (CIWA) • **thiamine before glucose** • lorazepam/oxazepam in liver disease.
3. **Delirium tremens** 48–96 h • seizures 24–48 h.
4. **Naltrexone** → CI with opioids; hepatotoxicity • **acamprosate** → renal, OK in liver disease • **disulfiram** → no alcohol 2 weeks after.
5. **Opioid overdose → naloxone** (short half-life, re-sedation).
6. **Buprenorphine** → start in mild–moderate withdrawal (else precipitated withdrawal) • ceiling effect.
7. **Methadone** → OTP only for OUD, QT, overdose risk early, CYP3A4/2B6 inducers ↓ levels.
8. **Naltrexone ER** → 7–10 days opioid-free.
9. **Lofexidine/clonidine** → opioid withdrawal symptoms.
10. **Pregnancy OUD → methadone or buprenorphine**.
11. **Stimulants and cannabis** → no approved drugs • cocaine: avoid acute beta-blockers.

13. Rheumatoid Arthritis and Osteoarthritis

Predicted for your exam (likelihood: VERY HIGH). RA on MTX stations in 2018–19, 2020, 2022–23, 2025 (pregnancy planning, monitoring).

1. ★ MTX **15 mg once weekly** → 20–25 mg + folic acid 1 mg daily; SC if inadequate; add TNFi if not at target at 3 months.
2. ★ MTX monitoring: CBC, LFT, SCr q2–4 weeks × 3 months, then q8–12 weeks, then q12 weeks; avoid TMP-SMX and alcohol.
3. ★ Pregnancy planning: stop MTX ≥3 months before conception; leflunomide cholestyramine washout; HCQ/sulfasalazine/certolizumab safe.
4. ★ Before biologics: TB (IGRA, CXR), HBV, HCV, HIV, vaccines; latent TB → treat ≥1 month before TNFi; active TB → hold TNFi.
5. OA: topical NSAID first for knee/hand; oral NSAID choice by CV/GI risk; acetaminophen limited benefit; max 3 g/day in elderly/liver disease.

Exam Traps and Frequently Tested Points

Rheumatoid arthritis

1. **RA:** symmetric small joints (MCP, PIP, wrists), **spares DIP**, morning stiffness **> 1 hour**, systemic features.
2. **Anti-CCP** = most specific antibody; RF less specific.
3. **Start DMARDs early** (within 3 months) • treat-to-target (remission/low activity) • reassess at 3 and 6 months.
4. **Methotrexate** = first-line (moderate–high activity) • **hydroxychloroquine** for low activity.
5. **MTX is ONCE WEEKLY** • target 20–25 mg/week • switch to **SC** before adding drugs • **folic acid 1 mg daily**.
6. **MTX:** hepatotoxicity, myelosuppression, pneumonitis, stomatitis • **CI: pregnancy** (stop 3 months before), liver disease, alcohol, CrCl < 30.
7. **MTX + TMP-SMX** → severe myelosuppression • NSAIDs, PPIs, penicillins ↑ MTX.
8. **Hydroxychloroquine:** retinopathy (eye exam baseline, yearly after 5 years), ≤ 5 mg/kg/day, safe in pregnancy, no lab monitoring.
9. **Sulfasalazine:** sulfa allergy, orange urine, G6PD hemolysis, agranulocytosis, oligospermia.
10. **Leflunomide:** hepatotoxic and teratogenic → **cholestyramine washout** (men and women).
11. **Triple therapy** = MTX + SSZ + HCQ.
12. **Not at target on MTX** → add biologic or JAK inhibitor.
13. **Steroids** = bridge only (low dose, taper) • **NSAIDs** = symptoms only, not disease-modifying.
14. **Never combine two biologics or a biologic + JAK inhibitor**.

15. **Before biologics/JAK: TB and hepatitis B** screening • no live vaccines • give Shingrix and inactivated vaccines first.
16. **TNF inhibitors CI: NYHA III–IV heart failure**, **MS/demyelinating disease**, active infection.
17. **Infliximab** → must be given with MTX (chimeric) • **certolizumab** → preferred in pregnancy.
18. **Rituximab** → infusion reactions, HBV reactivation, PML; premedicate • preferred with lymphoma history.
19. **Tocilizumab** → ↑ LFTs, ↑ lipids, **GI perforation**, masks CRP/fever.
20. **JAK inhibitors** → boxed: infection, death, cancer, MACE, thrombosis • after **TNF failure** • herpes zoster • ↑ LDL • tofacitinib + strong CYP3A4 inhibitor → 5 mg daily.
21. **Pregnancy:** avoid MTX, leflunomide, JAK inhibitors • OK: HCQ, SSZ, TNF inhibitors (certolizumab), low-dose prednisone.

Osteoarthritis

1. **OA:** weight-bearing joints, DIP (Heberden's) and PIP (Bouchard's), stiffness **< 30 min**, pain with activity, normal labs.
2. **Obesity** = most modifiable risk factor • **exercise and weight loss** = cornerstone.
3. **Cane in the opposite hand** of the affected knee/hip.
4. **Topical NSAIDs** → first-line for **knee** OA, especially elderly (≥ 75) and comorbidities • max 32 g/day diclofenac gel.
5. **Oral NSAIDs** = most effective oral drugs • lowest dose, shortest time.
6. **GI risk** → PPI or celecoxib (+ PPI if highest risk) • **CV risk** → naproxen lowest • avoid NSAIDs in CKD, HF, ≥ 20 weeks pregnancy.
7. **Ibuprofen blocks aspirin** → aspirin 30 min before or ibuprofen 8 h before.
8. **NSAID DDIs:** anticoagulants/SSRIs (bleeding), ACEi/ARB + diuretic (AKI), lithium, methotrexate.
9. **Acetaminophen** → limited benefit; max 3 g/day (2 g with liver disease/alcohol).
10. **Capsaicin** → knee (not hand); burning; 2–4 weeks.
11. **Duloxetine** → OA with widespread pain/depression • **tramadol** → if others fail (seizures, serotonin syndrome).
12. **Intra-articular steroids** → strongly recommended (knee, hip) • max every 3 months • hyperglycemia in diabetes.
13. **Not recommended: glucosamine**, chondroitin (knee/hip), hyaluronic acid, non-tramadol opioids, PRP, stem cells, MTX, HCQ, biologics.
14. **No drug slows OA progression** • severe disease → joint replacement.

14. Fluids and Electrolytes

Predicted for your exam (likelihood: HIGH). Embedded in DKA, sepsis, HF (K 5.6), TPN and QT cases.

1. ★ DKA potassium rules (hold insulin if K <3.3; add 20–30 mEq/L if 3.3–5.2).
2. ★ Sepsis: 30 mL/kg balanced crystalloid; albumin not first-line. Contrast prophylaxis with isotonic saline.
3. ★ K 5.6 in HF: hold ACEi/MRA, recheck; K >5.5 on MRA → halve or stop.
4. ★ QT drugs: keep K ≥4 and Mg ≥2. Amphotericin and aminoglycosides waste K and Mg.
5. Corrected Ca = Ca + 0.8 × (4 − albumin); K-phos: 1 mL = 3 mmol P + 4.4 mEq K; hyponatremia from SSRIs/SNRIs (SIADH), thiazides.

Exam Traps and Frequently Tested Points

1. **TBW = 60% (men), 50% (women)** • 2/3 ICF, 1/3 ECF • maintenance **30–35 mL/kg/day** or **4-2-1** rule.
2. **NS: Na 154, Cl 154, 308 mOsm** → hyperchloremic acidosis • **LR: Na 130, K 4, Ca 3, lactate 28** • balanced crystalloids preferred in most ICU patients.
3. **NS preferred in brain injury** and hypochloremic alkalosis • **no albumin in TBI** • **avoid hydroxyethyl starch**.
4. 1 L crystalloid → **~250 mL** stays intravascular • 1 L D5W → **~80 mL**.
5. **Sepsis: 30 mL/kg** crystalloid in 3 h.

6. **Serum osm = 2Na + glucose/18 + BUN/2.8** (275–295) • **osmolal gap > 10** → methanol, ethylene glycol, propylene glycol, mannitol.
7. **Corrected Na** = Na + 1.6 per 100 mg/dL glucose above 100.
8. **Peripheral line osmolarity ≤ 900 mOsm/L**.
9. **Never give sterile water IV** (hemolysis) • hypotonic fluids → hospital-acquired hyponatremia (children, post-op).
10. **Hyponatremia:** check osmolality → volume status → urine Na/osm.
11. **Thiazides** → hyponatremia • **SIADH drugs:** SSRIs, carbamazepine/oxcarbazepine, cyclophosphamide, vincristine, chlorpropamide, MDMA, desmopressin.
12. **SIADH:** euvolemic, urine osm > 100, urine Na > 30 → **fluid restriction** first.
13. **Severe symptomatic hyponatremia → 3% NaCl 100–150 mL bolus** (↑ Na 4–6 mEq/L).
14. **Correction limit: ≤ 8–10 mEq/L per 24 h** (≤ 8 if high ODS risk: Na ≤ 105, alcoholism, malnutrition, hypokalemia, liver disease).
15. **Overcorrection → D5W ± desmopressin • ODS** = dysarthria, dysphagia, paralysis, locked-in.
16. **Tolvaptan:** start in hospital, no fluid restriction 24 h, avoid liver disease, ≤ 30 days • **conivaptan** = strong CYP3A4 inhibitor.
17. **Hypernatremia:** free water deficit = **TBW × (Na/140 − 1)** • correct ≤ **10–12 mEq/L/24 h** • shock → NS first.
18. **Lithium, demeclocycline, amphotericin** → nephrogenic DI • central DI → **desmopressin** • nephrogenic → thiazide, amiloride (lithium).
19. **Hypokalemia:** correct **Mg first** • ECG U waves, flat T • 10 mEq KCl ↑ K ~0.1.
20. **IV KCl: peripheral ≤ 10 mEq/h, central ≤ 20 mEq/h**, **never IV push**, avoid dextrose diluent.
21. **Hyperkalemia drugs:** ACEi/ARB, spironolactone, trimethoprim, NSAIDs, heparin, tacrolimus, cyclosporine, succinylcholine, digoxin toxicity.
22. **Hyperkalemia steps: calcium gluconate** (stabilize, doesn't lower K) → **insulin + dextrose**, albuterol, (bicarbonate if acidosis) → **remove**: loop diuretic, SZC, patiromer, SPS, dialysis.
23. **Peaked T waves** = hyperkalemia • rule out **hemolyzed sample**.
24. **SZC** → fastest binder (1 h), edema • **patiromer** → hypomagnesemia, separate 3 h • **SPS** → intestinal necrosis.
25. **1 g MgSO4 = 8.1 mEq = 4 mmol** • infuse slowly (≤ 1 g/h) • torsades → 1–2 g IV.
26. **Hypomagnesemia drugs: PPIs**, diuretics, cisplatin, amphotericin, aminoglycosides, cyclosporine/tacrolimus, cetuximab.
27. **Hypermagnesemia:** first sign **loss of deep tendon reflexes** → antidote **calcium gluconate**.
28. **Hypophosphatemia: refeeding syndrome**, alcoholism, DKA treatment, binders, IV ferric carboxymaltose → weakness, **respiratory failure**, hemolysis.
29. **K phosphate** if K < 4 (4.4 mEq K per 3 mmol) • **Na phosphate** if K ≥ 4 • dose in **mmol**.
30. **Corrected Ca = Ca + 0.8 × (4 − albumin)** • ionized Ca best in ICU.
31. **Hypocalcemia:** Chvostek, Trousseau, tetany, ↑ QT • **calcium gluconate IV** (chloride = 3× Ca, central, vesicant) • correct Mg.
32. **Calcium carbonate** 40% with meals (PPIs ↓) • **calcium citrate** 21%, any time.
33. **Hypercalcemia:** hyperparathyroidism and malignancy > 90% • stones, bones, groans, moans • short QT.
34. **Hypercalcemia treatment: NS first** → **calcitonin** (fast, tachyphylaxis) + **zoledronic acid** (2–4 days) • steroids for lymphoma/sarcoid • loop diuretics only after rehydration.
35. **Thiazides, lithium, vitamin D/A, calcium + antacids (milk-alkali)** → hypercalcemia.

15. Nutrition (EN/PN)

Predicted for your exam (likelihood: VERY HIGH). TPN calculations in Crohn's (2022–23, 2025), HF (2024) and critical care.

1. ★ Calculate total kcal (kg × kcal/kg), protein (g/kg × 4 kcal), dextrose (g × 3.4 kcal), lipid (20% = 2 kcal/mL), GIR (g × 1000/kg/1440; keep ≤4–5).
2. ★ Refeeding: thiamine 100 mg before dextrose, start at ~50% of goal, correct K, Mg, phosphate.

3. ★ Hold lipid if TG >400; limit ≤1 g/kg/day. Peripheral PN ≤900 mOsm/L.
4. ★ Protein: CKD 0.6–0.8, HD 1.2–1.5, critical illness 1.2–2.0 g/kg. HF → concentrated PN, fluid restriction.
5. When to start PN: NPO >7 days with low risk (sooner if malnourished); EN preferred whenever the gut works. Insulin in bag 0.05–0.1 unit/g dextrose.

Exam Traps and Frequently Tested Points

1. **"If the gut works, use it"** → EN preferred over PN • start EN within **24–48 h** in critically ill stable patients.
2. **Albumin and prealbumin** are negative acute-phase reactants → **not** reliable nutrition markers.
3. **Energy 25–30 kcal/kg/day** • ICU first week 12–25 • **protein 1.2–2 g/kg** (ICU) • 0.8 healthy • obese ICU: hypocaloric high-protein.
4. **Dextrose 3.4 kcal/g** • protein 4 • fat 9 • **lipid 20% = 2 kcal/mL** (10% = 1.1, 30% = 3) • **propofol 1.1 kcal/mL**.
5. **Nitrogen balance** = protein/6.25 − (UUN + 4).
6. **EN contraindications:** obstruction, severe ileus, high-output fistula, GI bleed, ischemia, shock.
7. **NG/NJ** short term (< 4–6 weeks) • **PEG/J-tube** long term • **post-pyloric = continuous only** • confirm placement by X-ray.
8. **Aspiration prevention:** head of bed **30–45°**, post-pyloric, prokinetics • don't routinely check GRV; don't hold for GRV < 500 mL.
9. **Diarrhea on EN → check sorbitol-containing meds, Mg, antibiotics, C. diff first**.
10. **Calorie-dense (1.5–2 kcal/mL)** for fluid restriction • renal formula: low K/phos • elemental for malabsorption • no pulmonary formula.
11. **Flush 30 mL water** before/after meds and every 4 h • unclog with warm water or pancreatic enzyme + bicarbonate (**not cola**).
12. **Don't crush EC, ER, SL, hazardous drugs** • give drugs separately • never in the feed bag.
13. **Phenytoin + tube feeds** → hold 1–2 h before/after • **warfarin** → INR changes • **quinolones** → chelation; no cipro suspension via tube • **levothyroxine** → separate.
14. **PN timing:** well-nourished low risk → wait 7 days • high risk/malnourished → start ASAP • supplemental PN after 7–10 days if EN < 60%.
15. **PPN ≤ 900 mOsm/L**, short term • CPN for full needs.
16. **Osmolarity:** dextrose 5 mOsm/g, amino acid 10 mOsm/g, electrolytes 2 mOsm/mEq.
17. **2-in-1 → 0.22-µm filter • TNA → 1.2-µm filter** • lipid alone hangs 12 h, TNA 24 h.
18. **Max GIR 4–5 mg/kg/min** • overfeeding dextrose → hyperglycemia, steatosis, ↑ CO2 (ventilator weaning).
19. **Regular insulin** is the only insulin added to PN • glucose goal 140–180.
20. **Hold lipids if TG > 400** • lipid CI: egg/soy (fish) allergy • EFAD: scaly rash, triene:tetraene > 0.2.
21. **Soybean lipid** (phytosterols) → PNALD/cholestasis → limit ≤ 1 g/kg/day, switch to SMOF/fish oil.
22. **Ca-phos precipitation** → fatal emboli • use **calcium gluconate**, add **phosphate first, calcium last** • ↑ risk with high pH, high temperature, low amino acids.
23. **Never add bicarbonate to PN** → use **acetate** for acidosis.
24. **Multivitamin has vitamin K** → warfarin • **↓ copper and manganese in cholestasis** • ↑ zinc with diarrhea/fistula.
25. **PN monitoring:** glucose q6h initially • electrolytes daily until stable • TG baseline and weekly • LFTs weekly.
26. **Catheter occlusion:** clot → **alteplase 2 mg** • lipid → **70% ethanol** • Ca-phos → **0.1 N HCl**.
27. **Most common PN metabolic complication: hyperglycemia** • most serious catheter complication: **CLABSI** (Candida).
28. **Abrupt PN stop** → hypoglycemia → taper.
29. **Refeeding syndrome:** ↓ **phosphate** (hallmark), ↓ K, ↓ Mg, thiamine deficiency • first 72 h • start **10–20 kcal/kg**, advance over 3–7 days, **thiamine 100 mg before feeding**, monitor electrolytes q12h × 3 days.
30. **Refeeding risk:** BMI < 16, weight loss > 15%, no intake > 10 days, low baseline electrolytes, alcoholism, anorexia.

16. Thyroid, Osteoporosis and Obesity

Predicted for your exam (likelihood: VERY HIGH). Thyroid storm/myxedema 2018–19, 2020, 2021; hypothyroid dosing yearly; amiodarone thyrotoxicosis 2025.

1. ★ Thyroid storm sequence: PTU → iodine ≥1 h later → propranolol → hydrocortisone → acetaminophen (not aspirin).
2. ★ Myxedema coma: hydrocortisone first, then LT4 200–400 mcg IV.
3. ★ Levothyroxine: check adherence/administration first, +12.5–25 mcg, TSH 6–8 weeks; pregnancy +25–30%; rifampin and iron/calcium interactions.
4. ★ Graves: methimazole (PTU in first trimester), agranulocytosis counseling; amiodarone-induced thyrotoxicosis type 1 vs 2.
5. Osteoporosis: fracture on alendronate → zoledronic acid/denosumab/teriparatide; alendronate counseling; romosozumab not within 1 year of MI.
6. Obesity with prediabetes/HTN: semaglutide 2.4 mg weekly or tirzepatide; screen MTC/MEN2.

Exam Traps and Frequently Tested Points

1. **TSH = best screening test** for primary thyroid disease • **central hypothyroidism → follow free T4**, not TSH.
2. **Hashimoto's → anti-TPO • Graves' → TSI/TRAb • RAIU high** in Graves'/toxic nodules, **low** in thyroiditis and exogenous hormone.
3. **Levothyroxine = 1.6 mcg/kg/day • elderly/CAD start 12.5–25 mcg** • check TSH every **4–8 weeks** • half-life 7 days.
4. Levothyroxine on an **empty stomach 30–60 min before breakfast • separate 4 h** from Ca, Fe, antacids, sucralfate, bile acid sequestrants, orlistat.
5. **Pregnancy: ↑ levothyroxine ~30% (2 extra doses/week)** immediately • goal TSH < 2.5.
6. Estrogen and pregnancy ↑ **TBG** → ↑ dose • enzyme inducers ↑ dose • correcting hypothyroidism ↑ **warfarin** effect.
7. **Myxedema coma:** IV levothyroxine 200–400 mcg + **hydrocortisone 100 mg q8h first/together** • passive rewarming.
8. **Methimazole = preferred** thionamide • **PTU: 1st trimester and thyroid storm • PTU boxed hepatotoxicity** • MMI teratogenic.
9. **Agranulocytosis** with thionamides → **fever or sore throat → stop the drug and get a CBC**.
10. Thionamides take **4–8 weeks** to work (stored hormone) → **beta-blocker** for symptoms • propranolol ↓ T4→T3.
11. **Thyroid storm order: PTU → (≥ 1 h later) iodine → propranolol → hydrocortisone • acetaminophen, not aspirin**.
12. **RAI:** CI in pregnancy/breastfeeding • → hypothyroidism • can worsen ophthalmopathy (prednisone) • stop MMI 3–5 days before.
13. **Thyroiditis (postpartum, painless, subacute) → beta-blocker only** (no thionamides).
14. **Amiodarone:** hypo- or hyperthyroidism • **type 1 → thionamide • type 2 → prednisone** • TFTs every 3–6 months • **lithium** → hypothyroidism.
15. **Nonthyroidal illness** (↓ T3) → do not treat • **biotin** → false lab results.
16. **Osteoporosis: T-score ≤ −2.5** • osteopenia −1 to −2.5 • **hip/vertebral fragility fracture = osteoporosis** regardless of BMD.
17. **Z-score** for premenopausal women and men < 50 • screen **women ≥ 65**, men ≥ 70.
18. Treat osteopenia if **FRAX ≥ 20% major or ≥ 3% hip**.
19. **Calcium 1,200 mg/day** (women > 50) • **max 500–600 mg per dose • carbonate 40% with meals • citrate 21% any time (PPIs)**.
20. **Vitamin D 800–1,000 IU/day** • goal ≥ 30 ng/mL • correct hypocalcemia/vitamin D before bisphosphonates and denosumab.
21. **Glucocorticoids** = most common drug cause • also PPIs, anticonvulsants, heparin, **DMPA (boxed, 2 years)**, aromatase inhibitors, excess levothyroxine.

22. **Oral bisphosphonates:** morning, empty stomach, **full glass of plain water**, **upright 30 min** (ibandronate 60), no food/meds 30 min.
23. **Ibandronate and raloxifene → vertebral fractures only** • alendronate, risedronate, zoledronic acid, denosumab → hip too.
24. **Zoledronic acid 5 mg yearly** • CI CrCl < 35 • acute-phase reaction • **drug holiday** after 5 y oral / 3 y IV if not high risk.
25. **ONJ** (dental exam first) and **atypical femur fracture** (thigh pain) with bisphosphonates and denosumab.
26. **Denosumab 60 mg SC q6 months • no holiday — rebound vertebral fractures** → switch to bisphosphonate • **hypocalcemia (boxed in CKD)**.
27. **Very high risk → anabolic first** (teriparatide 20 mcg, abaloparatide 80 mcg, romosozumab) → then antiresorptive.
28. **Teriparatide:** orthostatic hypotension, hypercalcemia • avoid in Paget's, radiation, open epiphyses.
29. **Romosozumab:** 12 months only • **boxed MI/stroke** — not within 1 year of MI/stroke.
30. **Raloxifene:** ↓ breast cancer, **boxed VTE/stroke**, hot flashes • stop 72 h before immobilization • **calcitonin** = least effective, malignancy risk, analgesic.
31. **BMI ≥ 30 = obesity** (≥ 25 in Asians) • waist > 40 in men, > 35 in women • goal **5–10%** loss.
32. **Drug therapy: BMI ≥ 30 or ≥ 27 + comorbidity • stop if < 5% loss at 12–16 weeks** • all CI in pregnancy.
33. **Tirzepatide (~20%) > semaglutide (~15%)** > Qsymia > liraglutide/Contrave > orlistat • GLP-1s: **boxed MTC/MEN2**, pancreatitis, gallstones.
34. **Tirzepatide ↓ oral contraceptive efficacy** (4 weeks after start/increase) • stop semaglutide **2 months before pregnancy**.
35. **Phentermine:** C-IV, short term, CI CVD/HTN/hyperthyroidism/glaucoma/MAOIs • **Qsymia: REMS (oral clefts), taper off, paresthesia, kidney stones**.
36. **Contrave: CI opioids, seizures, eating disorders** • boxed suicidality • CYP2D6 inhibitor • no high-fat meals.
37. **Orlistat:** TID with fatty meals • oily spotting • **MVI at bedtime** • separate levothyroxine 4 h, cyclosporine 3 h • ↑ INR.
38. **Weight-gaining drugs:** olanzapine, clozapine, mirtazapine, paroxetine, valproate, gabapentin, insulin, SU, TZDs, steroids, DMPA → review first.
39. **Bariatric surgery:** BMI ≥ 35 (or 30 with T2DM) • avoid **ER products and NSAIDs** • lifelong B12, iron, calcium citrate, vitamin D.

17. Diabetes

Predicted for your exam (likelihood: VERY HIGH). Insulin stations every year 2018–2023; DKA 2018–2025; new T2DM with HTN 2020.

1. ★ Insulin TDD 0.4–0.6 units/kg, 50/50, ICR 500/TDD, CF 1800/TDD; NPH/Mixtard → basal-bolus with 10–20% reduction if hypoglycemia.
2. ★ DKA step-by-step and resolution criteria; SC overlap 1–2 h.
3. ★ T2DM drug choice by comorbidity: ASCVD/HF/CKD → SGLT2i or GLP-1 RA; avoid TZD/saxagliptin in HF; avoid SGLT2i with recurrent UTIs/after DKA.
4. ★ Pregnancy: switch to insulin; targets 95/140/120; stop ACEi/ARB/statin.
5. ★ Insulin counseling and hypoglycemia 15-15 rule; glucose-log pattern adjustment.
6. HHS: fluids priority, insulin 0.1 unit/kg/h only if K ≥3.3, dextrose at 300; statin for DM 40–75; BP <130/80.

Exam Traps and Frequently Tested Points

1. **Diagnosis:** A1C ≥ 6.5% • FPG ≥ 126 • 2-h OGTT ≥ 200 • random ≥ 200 + symptoms • **prediabetes 5.7–6.4 / 100–125 / 140–199** • needs 2 abnormal tests.
2. **A1C unreliable:** anemia, hemoglobinopathies, pregnancy, transfusion, EPO, dialysis • **A1C 7% = eAG 154 mg/dL**.

3. **Screen all adults ≥ 35** (or overweight + risk factors) every 3 years • prior GDM → every 3 years for life.
4. **Goals:** A1C < 7% • preprandial 80–130 • postprandial < 180 • **BP < 130/80 • TIR > 70%**, TBR < 4%.
5. **DPP lifestyle: 7% weight loss + 150 min/week** → ↓ 58% • metformin for prevention: BMI ≥ 35, < 60 years, prior GDM.
6. **ASCVD → GLP-1 RA or SGLT2i with proven benefit • HF → SGLT2i • CKD → SGLT2i (eGFR ≥ 20)** + finerenone + GLP-1 RA • obesity → tirzepatide/semaglutide.
7. **A1C ≥ 1.5% above goal → dual therapy • A1C > 10% or BG ≥ 300 or catabolic → insulin** • GLP-1 RA before insulin.
8. **Metformin:** CI eGFR < 30 • don't start 30–45 • **B12 deficiency** • GI → titrate with food • hold for contrast (restart 48 h).
9. **SGLT2i:** genital mycotic infections, UTI, volume depletion, **euglycemic DKA**, Fournier's • **canagliflozin → amputation/fractures** • hold 3 days pre-op (ertugliflozin 4).
10. **GLP-1 RA: boxed MTC/MEN2** • GI effects • pancreatitis • gallbladder • **Rybelsus: empty stomach, ≤ 4 oz water, 30 min** • tirzepatide ↓ OC efficacy.
11. **Never combine DPP-4i with GLP-1 RA • linagliptin: no renal adjustment • saxagliptin/alogliptin → HF** • DPP-4i: arthralgia, bullous pemphigoid.
12. **TZDs:** boxed HF (CI NYHA III–IV), edema, weight gain, fractures, bladder cancer (pio), slow onset 2–3 months • gemfibrozil → pio max 15 mg.
13. **Sulfonylureas:** hypoglycemia + weight gain • **glyburide avoid in elderly/CKD → glipizide preferred** • CYP2C9 inhibitors (fluconazole, SMX/TMP) ↑ hypoglycemia.
14. **Repaglinide + gemfibrozil = CI** • glinides: skip meal, skip dose.
15. **Acarbose:** take with first bite • flatulence • **treat hypoglycemia with glucose, not sucrose**.
16. **Pramlintide:** ↓ prandial insulin by 50% • boxed severe hypoglycemia.
17. **Insulin kinetics:** rapid 15 min / 1–2 h / 3–5 h • regular 30 min / 2–3 h / 5–8 h • **NPH 1–2 h / 4–12 h / 12–18 h** • glargine flat ~24 h • degludec ~42 h.
18. **Clear before cloudy** (regular then NPH) • **never mix glargine/detemir/degludec • only regular (and rapid analogs) IV**.
19. **In-use insulin ~28 days** at room temperature (degludec/Toujeo 56) • unopened in fridge • rotate sites (lipohypertrophy).
20. **T1DM TDD 0.5 units/kg** (0.4–1) • **50% basal / 50% bolus • ICR 500/TDD • ISF 1800/TDD**.
21. **T2DM basal start 10 units or 0.1–0.2 units/kg** • ↑ 2 units every 3 days • hypoglycemia → ↓ 10–20% • **> 0.5 units/kg = overbasalization** → add GLP-1 or prandial.
22. **NPH BID → glargine = 80%** of total NPH • prandial start 4 units or 10% of basal at largest meal.
23. **Hypoglycemia:** level 1 < 70 • **level 2 < 54** • level 3 needs assistance • **Rule of 15** (4 glucose tabs / 4 oz juice) • glucagon 1 mg or nasal 3 mg.
24. **Beta-blockers mask hypoglycemia** except **sweating** • hypoglycemia unawareness → relax targets for weeks.
25. **DKA:** glucose ≥ 200, **pH < 7.3 / HCO3 < 18**, ketones ≥ 3, anion gap • **HHS:** glucose ≥ 600, **osm > 300**, no significant ketosis, older T2DM.
26. **DKA order: fluids first** → **check K** (**< 3.5 hold insulin**) → **regular insulin 0.1 units/kg/h** → **dextrose when BG < 250** (HHS < 300) → bicarb only pH < 7.0.
27. **DKA resolved:** ketones < 0.6, pH ≥ 7.3 or HCO3 ≥ 18 • **give SC basal 1–2 h before stopping the drip**.
28. **SGLT2i → euglycemic DKA** (glucose may be < 250) • children → cerebral edema.
29. **Statins:** all 40–75 (moderate) • high-intensity if risk factors or ASCVD • **ASCVD LDL < 55** • aspirin 75–162 mg for secondary prevention.
30. **UACR yearly** (T1 after 5 years, T2 at diagnosis) • **≥ 30 mg/g → ACEi/ARB** • SGLT2i eGFR ≥ 20 • finerenone (K) • **never ACEi + ARB**.
31. **Eye exam:** T1 within 5 years, T2 at diagnosis, then every 1–2 years • anti-VEGF • pregnancy → each trimester.
32. **Neuropathy:** monofilament yearly • **pregabalin, duloxetine**, TCAs, Na channel blockers • avoid opioids • check B12 (metformin).
33. **Gastroparesis:** metoclopramide ≤ 12 weeks (TD boxed) • stop GLP-1 RA/pramlintide • **ED: PDE5i — CI nitrates**.

34. **Feet:** daily self-inspection, no barefoot, no lotion between toes • comprehensive exam yearly.
35. **GDM:** screen 24–28 weeks • one-step **92 / 180 / 153** • two-step 50-g ≥ 130–140 → 100-g (95/180/155/140, ≥ 2 values).
36. **Pregnancy goals: fasting < 95**, **1-h < 140**, **2-h < 120**, A1C < 6% • **insulin preferred** • metformin and glyburide cross placenta • **CI ACEi/ARB/statins** • aspirin from 12–16 weeks.
37. **Postpartum GDM → 75-g OGTT at 4–12 weeks** • lifelong screening every 1–3 years • insulin needs fall sharply after delivery.
38. **Hospital:** target 140–180 • **basal-bolus**, **avoid sliding scale alone** (Beers) • ICU → IV insulin • steroid hyperglycemia → **NPH with the morning steroid**.
39. **Sick days:** never stop basal insulin (T1) • check ketones • hold **SADMANS** (SU, ACEi, diuretics, metformin, ARBs, NSAIDs, SGLT2i).
40. **Drugs ↑ glucose:** steroids, olanzapine/clozapine, thiazides, tacrolimus, protease inhibitors, niacin, statins, ICIs • **fluoroquinolones and pentamidine → both hypo and hyper**.
41. **Elderly:** avoid glyburide/glimepiride and sliding scale alone • A1C < 7–7.5% (healthy) / < 8% (complex) • deintensify.
42. **Vaccines:** influenza, pneumococcal, **hepatitis B (19–59)**, COVID, zoster ≥ 50, RSV (older adults).

18. Acute Medical Diseases (PAD, DKA, sepsis, shock, arrest, acid-base)

Predicted for your exam (likelihood: VERY HIGH). Septic shock 2018–19, 2020; DKA; anaphylaxis 2020–21; hemorrhagic shock and acid-base in critical care.

1. ★ Septic shock: hour-1 bundle, 30 mL/kg, NE → vasopressin 0.03 at NE ≥0.25 → epinephrine; hydrocortisone 50 q6h; MAP ≥65.
2. ★ Anaphylaxis: epinephrine 0.3–0.5 mg IM thigh q5–15 min; glucagon if on β-blocker; observe 4–6 h.
3. ★ Acid-base: AG, Winter's formula, delta ratio; metabolic alkalosis urine Cl <25 = saline-responsive.
4. ★ Hemorrhagic shock: 1:1:1, TXA 1 g + 1 g over 8 h within 3 h, calcium, permissive SBP 80–90.
5. Bradycardia: atropine 1 mg (max 3) → dopamine/epinephrine/pacing; HTN emergency ≤25% first hour.
6. ICU prophylaxis: SUP (ventilated >48 h), VTE (enoxaparin 40, UFH if CrCl <30), glucose 140–180.

Exam Traps and Frequently Tested Points

1. **Treat pain first** (analgesia-first) • **NRS** if able; **BPS > 5** or **CPOT > 2** if not • vital signs alone are not valid.
2. **Fentanyl**: no histamine release, safe in renal failure, chest wall rigidity • **morphine**: histamine, avoid in renal failure • **remifentanil**: esterase • **avoid meperidine**.
3. **RASS goal −2 to 0** (light sedation) • daily SAT + SBT • **propofol or dexmedetomidine over benzodiazepines**.
4. **Benzodiazepines preferred only for** alcohol/BZD withdrawal, seizures, deep sedation/amnesia with paralysis.
5. **Propofol:** hypotension, ↑ TG (check 48 h), pancreatitis, **1.1 kcal/mL**, **PRIS** (> 70 mcg/kg/min > 48 h → acidosis, rhabdo, hyperK, arrhythmia).
6. **Dexmedetomidine:** α2 agonist, no respiratory depression, arousable, bradycardia/hypotension, ↓ delirium, not for seizures.
7. **Lorazepam infusion → propylene glycol toxicity** (↑ osmolal gap, AG acidosis, AKI) • **midazolam**: CYP3A4, accumulates in renal failure.
8. **NMBs have no sedation or analgesia** • cisatracurium = Hofmann (organ independent) • succinylcholine → hyperkalemia, MH (dantrolene).
9. **CAM-ICU**: acute change/fluctuation + inattention + (altered LOC or disorganized thinking) • ICDSC ≥ 4 • **hypoactive delirium most common**.
10. **No drug for delirium prevention** • antipsychotics only short-term for distressing symptoms • **benzodiazepines worsen delirium** (except withdrawal) • haloperidol → QTc (hold > 500).
11. **ABCDEF bundle** • early mobility • sleep hygiene • reorientation, glasses, hearing aids.

12. **DKA:** glucose ≥ 200, **pH < 7.3 / HCO3 < 18**, **β-OHB ≥ 3** • **HHS:** glucose ≥ 600, effective osm > 300, minimal ketones.
13. **DKA order: fluids → check K** (< 3.5 → **hold insulin**) → **insulin 0.1 units/kg/h** → **dextrose at BG < 250** (HHS < 300) → bicarbonate only pH < 7.0 • phosphate only < 1.
14. DKA resolved: **β-OHB < 0.6 + pH ≥ 7.3 or HCO3 ≥ 18 • SC basal 1–2 h before stopping IV insulin** • K looks normal but total body K is depleted.
15. **Euglycemic DKA with SGLT2i • cerebral edema in children** • corrected Na = Na + 1.6 per 100 mg/dL glucose above 100.
16. **Sepsis = organ dysfunction (SOFA ≥ 2) • septic shock = vasopressor for MAP ≥ 65 + lactate > 2** despite fluids • **qSOFA not alone** for screening.
17. **Hour-1 bundle:** lactate, **blood cultures before antibiotics**, broad-spectrum antibiotics, **30 mL/kg crystalloid** (hypotension or lactate ≥ 4), vasopressors MAP ≥ 65.
18. **Antibiotics within 1 h** in shock/probable sepsis • MRSA coverage if risk • double gram-negative only if MDR risk • extended-infusion β-lactams • procalcitonin to stop.
19. **Balanced crystalloids > NS** • albumin after large volumes • **no starches** • dynamic measures (passive leg raise) to assess fluid responsiveness.
20. **Norepinephrine first → vasopressin 0.03 units/min** (NE 0.25–0.5) → epinephrine • dobutamine for cardiac dysfunction • dopamine only selected • peripheral start OK • phentolamine for extravasation.
21. **Hydrocortisone 200 mg/day** if NE ≥ 0.25 mcg/kg/min for ≥ 4 h • **no vitamin C**, no bicarbonate unless pH ≤ 7.2 + AKI.
22. **Sepsis supportive:** Hb < 7 transfusion • glucose 144–180 (start ≥ 180) • LMWH prophylaxis • SUP only if risk • ARDS 6 mL/kg PBW, plateau ≤ 30, prone.
23. **Shock profiles:** hypovolemic ↓CO ↓PCWP ↑SVR • **cardiogenic ↓CO ↑PCWP ↑SVR • distributive ↑CO ↓SVR** • obstructive ↓CO ↑SVR.
24. **Hemorrhagic shock:** 1:1:1 blood products, limit crystalloids, **TXA within 3 h**, calcium • **cardiogenic:** PCI, NE (not dopamine), dobutamine/milrinone, IABP.
25. **Anaphylaxis: epinephrine IM 0.3–0.5 mg thigh first** • antihistamines/steroids are adjuncts • β-blocker patients → **glucagon**.
26. **Neurogenic shock:** hypotension + **bradycardia** • NE, atropine • MAP 85–90.
27. **Milrinone:** PDE3, inodilator, hypotension, **renal adjustment** • **phenylephrine:** pure α1, reflex bradycardia • **epinephrine ↑ lactate**.
28. **CPR: 100–120/min**, **2–2.4 in**, full recoil, minimal pauses, **30:2** or **1 breath q6s** with advanced airway • **ETCO2 < 10** = poor CPR.
29. **VF/pVT:** shock → CPR → shock → **epinephrine 1 mg q3–5 min** → shock → **amiodarone 300 → 150 mg** or lidocaine 1–1.5 mg/kg.
30. **PEA/asystole: epinephrine ASAP**, no shock, **no atropine** • look for **H's and T's**.
31. **Torsades → magnesium 1–2 g • hyperkalemia → calcium chloride 1 g** + bicarbonate + insulin/dextrose • PE arrest → alteplase • LAST → lipid emulsion.
32. **Vasopressin, bicarbonate, calcium not routine** in arrest • IV first, IO second • ET route NAVEL (2–2.5× dose).
33. **Bradycardia: atropine 1 mg q3–5 min (max 3 mg)** → pacing / dopamine / epinephrine • **SVT:** vagal → **adenosine 6 → 12 mg** rapid push (↓ with dipyridamole/carbamazepine/central line; theophylline ↓ effect) • unstable → **synchronized cardioversion**.
34. **Post-ROSC:** SpO2 92–98%, PaCO2 35–45, **MAP ≥ 65 / SBP ≥ 90**, PCI for STEMI, **temperature 32–37.5 °C and prevent fever ≥ 72 h**, prognosticate ≥ 72 h.
35. **Acid–base:** pH 7.35–7.45, PaCO2 35–45, HCO3 22–26, **AG = Na − (Cl + HCO3)** 8–12 • **correct AG for albumin (+2.5 per 1 g/dL)**.
36. **Winter's formula: PaCO2 = 1.5 × HCO3 + 8 ± 2** • metabolic alkalosis: ↑ 0.7 per 1 HCO3 • respiratory acute 1 (acidosis) / 2 (alkalosis) • chronic 3.5 / 4–5 per 10 mmHg.
37. **Delta-delta** < 1 → + NAGMA • > 2 → + metabolic alkalosis • **osmolal gap > 10** → toxic alcohols (fomepizole).
38. **AGMA = GOLDMARK • type B lactic acidosis drugs:** metformin, linezolid, propofol, NRTIs, epinephrine, nitroprusside.
39. **NAGMA:** diarrhea, RTA, NS • **urine AG negative = GI**, positive = RTA • RTA 1/2 → hypoK (amphotericin, acetazolamide, topiramate, tenofovir) • **RTA 4 → hyperK** (ACEi/ARB, spironolactone, trimethoprim, heparin).

40. **Bicarbonate deficit = 0.5 × wt × (desired − measured)**, give half • ADRs: hypernatremia, hypokalemia, ↓ iCa, ↑ CO2 • not routine in lactic acidosis/DKA.
41. **Metabolic alkalosis:** urine Cl **< 20 = saline-responsive** (vomiting, NG, diuretics remote → NS + KCl, PPI) • **> 20 = saline-resistant** (hyperaldosteronism → spironolactone) • **acetazolamide** if volume overloaded • HCl 0.1 N (central) if severe.
42. **Respiratory acidosis:** opioids, BZDs, COPD, neuromuscular → ventilate (BiPAP), naloxone, O2 88–92% in COPD • **respiratory alkalosis:** anxiety, PE, sepsis, salicylates (early), pregnancy, liver failure.
43. **Salicylate toxicity = respiratory alkalosis + AG metabolic acidosis** → urine alkalinization (bicarbonate), dialysis.

19. DVT and Pulmonary Embolism

Predicted for your exam (likelihood: VERY HIGH). PE in young woman 2022–23 and 2024; pregnancy anti-Xa 2018–19 and 2022–23; cancer VTE; PE with PCOS 2025.

1. ★ Enoxaparin 1 mg/kg q12h (calc with patient weight, round); UFH 80/18 for massive PE or planned lysis; alteplase 100 mg over 2 h.
2. ★ Warfarin overlap ≥5 days and INR ≥2 × 24 h; INR-based dose adjustments (3.7, 1.3, 1.8, 4.5 examples).
3. ★ Pregnancy anti-Xa peak 4–6 h after 3rd–4th dose, target 0.6–1.0; 0.25 → increase 10–25%.
4. ★ Thrombophilia/APS → warfarin not DOAC; estrogen contraceptives contraindicated after VTE.
5. ★ Cancer VTE: DOAC or LMWH (LMWH/apixaban for GI/GU cancer); duration while cancer active; second VTE indefinite.
6. DOAC VTE doses: apixaban 10 BID × 7 d → 5 BID (2.5 BID after 6 months); rivaroxaban 15 BID × 21 d → 20 daily.

Exam Traps and Frequently Tested Points

1. **Virchow's triad:** stasis, endothelial injury, hypercoagulability • **factor V Leiden** = most common inherited thrombophilia.
2. **Drugs ↑ VTE:** estrogens (COCs, HRT), **tamoxifen, raloxifene**, testosterone, **ESAs**, **thalidomide/lenalidomide**, tofacitinib, L-asparaginase.
3. **Wells DVT ≤ 1 → D-dimer**; ≥ 2 → compression ultrasound • **Wells PE ≤ 4 → D-dimer**; > 4 → **CTPA**.
4. **D-dimer rules OUT** (high sensitivity, low specificity) • age-adjusted cut-off = **age × 10** (> 50 years) • not useful in high probability.
5. **PERC rule** (8 items, low probability only) → no testing if all negative.
6. **CTPA** = test of choice for PE • **V/Q scan** if CKD, contrast allergy, pregnancy • unstable → **bedside echo** (RV dysfunction).
7. **ECG: sinus tachycardia most common**; S1Q3T3 classic • ABG: hypoxemia + respiratory alkalosis • troponin/BNP for risk stratification.
8. **High-risk PE (shock/hypotension) → thrombolysis (alteplase 100 mg over 2 h)** • intermediate-high → anticoagulate + monitor (rescue lysis) • low risk (**sPESI 0**) → outpatient DOAC.
9. **Hold UFH during alteplase**, restart without bolus when aPTT < 2× ULN • know **absolute contraindications** (prior ICH, stroke < 3 months, active bleeding, dissection).
10. **DOACs preferred over warfarin** for most VTE • **apixaban/rivaroxaban need no lead-in** • **dabigatran/edoxaban need 5–10 days parenteral first**.
11. **Apixaban 10 mg BID × 7 days → 5 mg BID** (extended 2.5 mg BID) • **rivaroxaban 15 mg BID × 21 days → 20 mg daily with food** (extended 10 mg).
12. **Dabigatran 150 mg BID**: avoid CrCl < 30, dyspepsia, don't open capsules, original bottle • **edoxaban 60 mg** (30 mg if CrCl 15–50, ≤ 60 kg, P-gp inhibitors).
13. **Cancer VTE → apixaban/rivaroxaban/edoxaban** (or LMWH, especially GI/GU cancers) • **APS → warfarin** (no DOACs) • **pregnancy → LMWH** • **CrCl < 30 → UFH/warfarin** (apixaban possible).
14. **Duration:** provoked → **3 months** • unprovoked → 3 months then **extended** if bleeding risk low • cancer → while active • 2nd unprovoked → indefinite.

15. **Isolated distal DVT** → serial ultrasound 2 weeks if no severe symptoms • **subsegmental PE** without DVT → surveillance • **SVT ≥ 5 cm → fondaparinux 2.5 mg × 45 days**.
16. **Warfarin:** overlap with parenteral **≥ 5 days AND INR ≥ 2 for 24 h** • start 5 mg (2.5 elderly) • goal INR 2–3 • factors II, VII, IX, X, proteins C and S.
17. **Warfarin skin necrosis** (protein C deficiency, days 3–8) • **teratogenic** • consistent vitamin K diet.
18. **Warfarin ↑ INR:** amiodarone, fluconazole, metronidazole, SMX-TMP, fluoroquinolones • **↓ INR:** rifampin, carbamazepine, phenytoin, St John's wort, vitamin K.
19. **INR 4.5–10 no bleeding → hold, no vitamin K • INR > 10 → oral vitamin K 2.5–5 mg • major bleeding → 4F-PCC + vitamin K 5–10 mg IV**.
20. **Reversal:** dabigatran → **idarucizumab 5 g** • Xa inhibitors → andexanet alfa or **4F-PCC • UFH → protamine 1 mg/100 units** • LMWH partial (~60%) • fondaparinux none.
21. **UFH:** 80 units/kg → 18 units/kg/h • aPTT 1.5–2.5× • ADRs: HIT, osteoporosis, **hyperkalemia** • preferred in unstable PE, renal failure, possible procedures.
22. **Enoxaparin: 1 mg/kg q12h** (1 mg/kg daily if CrCl < 30) • prophylaxis 40 mg daily (30 mg if CrCl < 30) • anti-Xa in pregnancy/obesity/renal • **boxed spinal hematoma**.
23. **Fondaparinux:** weight-based 5/7.5/10 mg • **CI CrCl < 30** • no HIT cross-reactivity.
24. **DOAC DDIs:** avoid **strong 3A4 + P-gp inhibitors** (ketoconazole, ritonavir) and **inducers** (rifampin, carbamazepine, phenytoin, St John's wort) • NSAIDs/SSRIs/antiplatelets ↑ bleeding.
25. **DOACs contraindicated** with mechanical heart valves and moderate–severe mitral stenosis; avoid in APS, pregnancy, breastfeeding.
26. **HIT:** platelets ↓ > 50% at **5–10 days** (or < 24 h if exposure within 100 days) + thrombosis • **4T score** → PF4 ELISA → serotonin release assay.
27. **HIT treatment: stop all heparin (including LMWH and flushes)** → **argatroban** (renal failure) / bivalirudin / fondaparinux / DOAC • **no platelet transfusions • no warfarin until platelets ≥ 150,000** (overlap ≥ 5 days) • argatroban falsely ↑ INR.
28. **IVC filter** only if acute VTE + **absolute contraindication to anticoagulation** • retrievable • not routine with anticoagulation.
29. **Compression stockings not routinely** to prevent post-thrombotic syndrome • early ambulation encouraged • home treatment for DVT/low-risk PE.
30. **Medical inpatient prophylaxis: Padua ≥ 4 → enoxaparin 40 mg daily** or UFH 5,000 q8–12h • bleeding risk → IPC • **aspirin not adequate** • no routine extended prophylaxis.
31. **Ortho (hip/knee):** LMWH, apixaban 2.5 BID, rivaroxaban 10 daily, aspirin, fondaparinux, warfarin • **10–14 days up to 35 days**.
32. **Cancer outpatients Khorana ≥ 2** → apixaban 2.5 mg BID or rivaroxaban 10 mg • **myeloma on lenalidomide** → aspirin or LMWH.
33. **Pregnancy:** LMWH throughout + **≥ 6 weeks postpartum** (≥ 3 months total) • warfarin OK postpartum/breastfeeding • DOACs avoid.
34. **CTEPH** → persistent dyspnea 3–6 months after PE → V/Q scan • pulmonary endarterectomy, **riociguat** (CI with PDE5i/nitrates).
35. **Counseling:** signs of bleeding, avoid NSAIDs/aspirin, herbal products, rivaroxaban with food, dabigatran original bottle, never double doses.

20. ID 1: UTI, Respiratory Infections, TB, Brucellosis

Predicted for your exam (likelihood: VERY HIGH). TB stations every year; CAP/HAP/VAP 2018–2025; pyelonephritis in pregnancy 2024.

1. ★ TB: RIPE doses, pyridoxine, 6 vs 9 months, hepatotoxicity stop/rechallenge, toxicity by drug, renal dosing, isolation removal criteria.
2. ★ HIV + TB: ART timing by CD4, dolutegravir 50 BID, TDF not TAF, no cobicistat; latent TB after INH hepatitis → 4R.
3. ★ HAP/VAP: MRSA >20% and FQ resistance >10% thresholds; 7 days; carbapenem–valproate and linezolid–SSRI interactions.
4. ★ CAP: ceftriaxone + azithromycin/doxycycline; QTc >500 → doxycycline; ofloxacin anaphylaxis → no FQ; chelation with Ca/Fe.

5. ★ Pyelonephritis in pregnancy: ceftriaxone, avoid FQ/TMP-SMX/tetracyclines; ESBL → meropenem; suppressive cephalexin.
6. Brucellosis (lower probability): doxycycline 100 BID × 6 weeks + rifampin or streptomycin/gentamicin; pregnancy/children → TMP-SMX or rifampin regimens.

Exam Traps and Frequently Tested Points

1. **E. coli** causes most UTIs • **S. saprophyticus** in young women • **Proteus** → struvite stones and **nitrofurantoin-resistant** • nitrite negative with Enterococcus/S. saprophyticus.
2. **Uncomplicated cystitis: nitrofurantoin 100 mg BID × 5 days**, **TMP-SMX DS BID × 3 days** (resistance < 20%), **fosfomycin 3 g once** • FQs reserved.
3. **Nitrofurantoin and fosfomycin are NOT for pyelonephritis** • nitrofurantoin avoid CrCl < 30 • pulmonary, hepatic, neuropathy toxicity.
4. **Pyelonephritis:** cipro 500 BID × 7 d or levo 750 × 5–7 d (if FQ resistance ≤ 10%, else initial ceftriaxone) • TMP-SMX 14 d if susceptible • ESBL → carbapenem.
5. **ASB:** treat only in **pregnancy** and before **urologic procedures with mucosal bleeding** • NOT in elderly, diabetics, catheters.
6. **Pregnancy UTI:** nitrofurantoin, cephalexin, amoxicillin-clavulanate, fosfomycin • **avoid FQs, tetracyclines, TMP-SMX (1st trimester/term)** • pyelo → admit, IV ceftriaxone.
7. **Men:** 7 days • prostatitis: FQ or TMP-SMX 2–4 weeks (chronic 4–6 weeks FQ) • CAUTI: change catheter, 7 days.
8. **Phenazopyridine** ≤ 2 days, orange urine, G6PD hemolysis • **methenamine** needs acidic urine.
9. **TMP-SMX:** hyperkalemia, ↑ INR, ↑ SCr (secretion), SJS • **FQ:** boxed tendon/neuropathy/CNS/MG, QT, chelation, ciprofloxacin–tizanidine CI.
10. **GAS pharyngitis:** penicillin V or amoxicillin × **10 days** (benzathine 1.2 MU once) • allergy: cephalexin (mild) / clindamycin or azithromycin 5 d (severe) • **negative RADT in children → throat culture**.
11. **AOM: high-dose amoxicillin 80–90 mg/kg/day** • amox-clav if amoxicillin in 30 days or conjunctivitis • duration 10 d (< 2 y), 7 d (2–5 y), 5–7 d (≥ 6 y) • observation option.
12. **Sinusitis bacterial if:** ≥ 10 days, severe ≥ 3–4 days (fever ≥ 39 + purulence), double sickening • **amox-clav first-line** 5–7 days adults • **no macrolides/TMP-SMX**.
13. **Common cold and acute bronchitis → no antibiotics** • pertussis → azithromycin.
14. **Influenza:** oseltamivir 75 mg BID × 5 d (≤ 48 h, but any time if hospitalized/high risk) • CrCl 30–60: 30 mg BID • zanamivir → bronchospasm • baloxavir single dose, no dairy.
15. **CAP outpatient healthy:** amoxicillin 1 g TID, doxycycline, or macrolide (if resistance < 25%) • **comorbidities:** amox-clav/cephalosporin + macrolide/doxy, or resp FQ.
16. **CAP inpatient:** β-lactam + macrolide or resp FQ alone • **ICU:** β-lactam + macrolide or β-lactam + FQ • add MRSA/Pseudomonas only with specific risk factors.
17. **CAP duration ≥ 5 days** (stable 48–72 h) • no routine anaerobic coverage for aspiration • procalcitonin not to withhold antibiotics • **CURB-65**: 0–1 home, 2 admit, ≥ 3 ICU.
18. **β-lactams don't cover atypicals** (Mycoplasma, Chlamydophila, Legionella) → macrolide, doxycycline, FQ • Legionella → hyponatremia, diarrhea.
19. **HAP/VAP:** ≥ 48 h after admission/intubation • cover S. aureus + Pseudomonas • MRSA (vanco/linezolid) and double antipseudomonal only with risk factors • **7 days**.
20. **Daptomycin never for pneumonia** (surfactant) • **moxifloxacin not for UTI** • **linezolid**: thrombocytopenia, serotonin syndrome, tyramine.
21. **TST cut-offs: ≥ 5 mm** HIV, contacts, transplant, immunosuppressed • **≥ 10 mm** immigrants, IDU, HCW, < 4 years, diabetes • **≥ 15 mm** no risk.
22. **TST read 48–72 h (induration)** • **IGRA preferred if BCG-vaccinated** • two-step TST for HCW baseline • positive test → **CXR** to rule out active TB.
23. **LTBI: 3HP** (INH + rifapentine weekly × 12), **4R**, **3HR** (preferred) • 6–9H alternative • **pyridoxine** with INH.
24. **Active TB: RIPE × 2 months → INH + RIF × 4 months** (6 months total) • **9 months** if cavitation + positive culture at 2 months • stop EMB when susceptible • DOT.
25. **Sputum × 3** AFB smear + culture (gold standard) + NAAT • airborne isolation, negative pressure, N95 • 3 negative smears to discontinue.

26. **INH:** hepatitis, **neuropathy (B6)**, lupus, ↑ phenytoin • **RIF: orange fluids (contact lenses)**, potent inducer (OCs, warfarin, PIs, azoles, DOACs, methadone).
27. **PZA:** hepatotoxicity (most), **hyperuricemia/gout** • **EMB: optic neuritis** (visual acuity, **red-green**) → monthly vision checks • PZA/EMB 3×/week in dialysis.
28. **Hepatotoxicity:** stop if ALT > 3× with symptoms or > 5× without → reintroduce RIF → INH → PZA last.
29. **TB in pregnancy:** INH + RIF + EMB × 9 months (no PZA in US) • avoid streptomycin, rifapentine • **HIV:** rifabutin with PIs, dolutegravir 50 mg BID with rifampin, ART within 2 weeks if CD4 < 50 (delay in TB meningitis).
30. **TB meningitis:** 9–12 months + dexamethasone • **MDR-TB** (INH + RIF resistant) → **BPaLM × 6 months** • bedaquiline QT.
31. **Brucellosis:** B. melitensis (goats/sheep/camels) • **raw milk** • undulant fever, **malodorous sweats**, arthralgia, hepatosplenomegaly • **sacroiliitis/spondylitis** most common complication • endocarditis = main cause of death.
32. **Brucellosis diagnosis:** blood culture (alert lab — lab-acquired infection) • SAT ≥ 1:160 (≥ 1:320 endemic) • 2-ME, Coombs • PCR.
33. **Brucellosis treatment: doxycycline × 6 weeks + streptomycin 2–3 weeks** (best) or **+ gentamicin 7 days** or **+ rifampin × 6 weeks • never monotherapy**.
34. **Spondylitis ≥ 12 weeks** (doxy + rif + aminoglycoside) • **neurobrucellosis:** ceftriaxone + doxy + rif 3–6 months • **endocarditis:** aminoglycoside + doxy + rif + surgery.
35. **Brucellosis in children < 8:** TMP-SMX + rifampin • **pregnancy:** rifampin ± TMP-SMX (avoid doxy, streptomycin) • relapse → repeat regimen • lab PEP: doxy + rif × 3 weeks.

21. ID 2: Endocarditis, SSTI/Osteomyelitis, CNS, IAI, C. difficile

Predicted for your exam (likelihood: VERY HIGH). Meningitis 2018–19 and 2022–23; endocarditis with gentamicin; MRSA osteomyelitis; peritonitis + CDI 2025.

1. ★ Meningitis empiric regimen with doses + dexamethasone timing; age-based organisms; CSF interpretation; chemoprophylaxis options.
2. ★ MSSA IE: nafcillin/cefazolin; prosthetic valve adds rifampin + gentamicin 2 weeks; gentamicin synergy trough <1.
3. ★ MRSA osteomyelitis on vancomycin with low GFR: AUC 400–600, what to do with level >20 or <10, when to repeat a level.
4. ★ Post-op peritonitis: pip-tazo or meropenem, 4 days after source control; Candida krusei → echinocandin.
5. ★ Fulminant CDI: vancomycin 500 mg q6h PO/NG + IV metronidazole ± rectal vancomycin; fidaxomicin for non-severe.
6. SSTI: purulent → I&D ± TMP-SMX/doxycycline; non-purulent → cephalexin; diabetic foot by severity.

Exam Traps and Frequently Tested Points

1. **IE organisms: S. aureus** most common (IDU → tricuspid) • viridans strep (dental) • enterococci (GU) • CoNS (early PVE) • **S. gallolyticus → colonoscopy** • HACEK.
2. **Osler nodes = painful** (immunologic) • **Janeway lesions = painless** (embolic) • Roth spots, splinter hemorrhages, glomerulonephritis.
3. **Duke definite:** 2 major, 1 major + 3 minor, or 5 minor • **3 sets of blood cultures before antibiotics** • TTE first → TEE (PVE, S. aureus, abscess).
4. **Duration from first negative blood culture** • NVE ~4 weeks (strep) / 6 weeks (staph, enterococcus) • PVE ≥ 6 weeks.
5. **Strep IE:** penicillin G or ceftriaxone 2 g × 4 weeks (or 2 weeks + gentamicin) • vancomycin if allergic.
6. **MSSA IE:** nafcillin/oxacillin (or cefazolin) × 6 weeks • **no gentamicin for NVE** • **MRSA:** vancomycin or daptomycin ≥ 8 mg/kg • β-lactam beats vancomycin for MSSA.
7. **PVE staph: add rifampin (biofilm) + gentamicin first 2 weeks** • start rifampin after bacteremia clears.
8. **Enterococcal IE: ampicillin + ceftriaxone** (CrCl < 50 or HLAR) or ampicillin + gentamicin • **cephalosporins alone don't cover enterococci**.

9. **Gentamicin synergy 3 mg/kg/day** (peak 3–4, trough < 1) • **daptomycin**: CPK weekly, not for pneumonia • nafcillin: interstitial nephritis, ↓ warfarin.
10. **Surgery:** HF, abscess/heart block, persistent bacteremia, fungal, vegetation > 10 mm with emboli • no aspirin for IE.
11. **IE prophylaxis:** only highest-risk (prosthetic valve, prior IE, certain CHD, transplant valvulopathy) + **dental procedures** • **amoxicillin 2 g** 30–60 min before • allergy: azithromycin 500 mg, doxycycline, cephalexin • **no clindamycin** • not for GI/GU.
12. **SSTI: purulent = S. aureus** (I&D ± TMP-SMX/doxy) • **non-purulent = strep** (cephalexin, penicillin, dicloxacillin, clindamycin × 5 days) • elevate limb, treat tinea pedis.
13. **TMP-SMX and doxycycline cover MRSA but poor/unreliable strep • cephalexin/dicloxacillin don't cover MRSA** • clindamycin covers both (resistance; C. diff).
14. **Necrotizing fasciitis:** pain out of proportion → **surgery** + vancomycin + pip-tazo + **clindamycin (toxin)** • GAS → penicillin + clindamycin • Vibrio → doxy + ceftazidime.
15. **Bites: amoxicillin-clavulanate** (Pasteurella — cats; Eikenella — humans) • avoid cephalexin/dicloxacillin/clindamycin alone • tetanus, rabies.
16. **Diabetic foot:** mild → gram-positive (cephalexin, amox-clav; MRSA → doxy/TMP-SMX) 1–2 weeks • moderate–severe → polymicrobial (amox-clav, ertapenem, pip-tazo if Pseudomonas risk) • osteomyelitis 6 weeks • deep tissue cultures, offloading.
17. **Osteomyelitis:** S. aureus most common • **Salmonella (sickle cell)** • **Pseudomonas (sneaker puncture)** • MRI + bone biopsy • **6 weeks** • oral FQs, linezolid, TMP-SMX OK.
18. **Septic arthritis:** arthrocentesis • S. aureus → vancomycin • gonococcal → ceftriaxone 1 g daily • rifampin combination for prosthetic hardware.
19. **Meningitis triad** fever + neck stiffness + altered mental status • **antibiotics within 1 h, don't wait for CT/LP** • CT first if immunocompromised, seizure, papilledema, focal deficit, ↓ consciousness.
20. **CSF bacterial:** ↑ PMNs, **↓ glucose (< 40)**, ↑↑ protein • viral: lymphocytes, normal glucose.
21. **Empiric meningitis:** < 1 month **ampicillin + cefotaxime (no ceftriaxone in neonates)** • 1 month–50 years **vancomycin + ceftriaxone** • **> 50 years/immunocompromised + ampicillin (Listeria)** • post-neurosurgery vancomycin + cefepime/meropenem.
22. **Ceftriaxone 2 g q12h** (meningitic dose) • **ampicillin 2 g q4h** • **cephalosporins don't cover Listeria** (TMP-SMX in PCN allergy).
23. **Dexamethasone 0.15 mg/kg q6h × 2–4 days before/with first antibiotic** → benefit in pneumococcal (adults) and Hib (children) • stop if other organism • ↓ vanco CSF penetration.
24. **Duration:** meningococcus/Hib 7 days • pneumococcus 10–14 • GBS 14–21 • **Listeria ≥ 21** • gram-negatives 21 days.
25. **Meningococcal prophylaxis: rifampin 600 mg q12h × 2 days**, **ciprofloxacin 500 mg once**, or **ceftriaxone 250 mg IM once (pregnancy)** • droplet precautions 24 h • Hib prophylaxis rifampin × 4 days.
26. **HSV encephalitis:** temporal lobe • **acyclovir 10 mg/kg q8h (IBW) × 14–21 days** • hydrate (crystal nephropathy) • cryptococcal: L-AmB + flucytosine → fluconazole, serial LPs.
27. **IAI = source control + antibiotics** • CA mild–moderate: **ceftriaxone/cefazolin + metronidazole**, ertapenem, FQ + metronidazole • severe/HA: **pip-tazo, carbapenem, cefepime + metronidazole**.
28. **Avoid ampicillin-sulbactam (E. coli), cefotetan and clindamycin (B. fragilis)** for IAI • no routine antifungal/enterococcal coverage in community-acquired.
29. **IAI duration 4 days after source control** (STOP-IT) • appendectomy for uncomplicated appendicitis → ≤ 24 h • uncomplicated diverticulitis may not need antibiotics.
30. **SBP:** cefotaxime 2 g q8h or ceftriaxone × 5–7 d + **albumin 1.5 g/kg day 1, 1 g/kg day 3** • prophylaxis cipro or TMP-SMX • necrotizing pancreatitis: no prophylactic antibiotics.
31. **Ertapenem** lacks Pseudomonas, Acinetobacter, enterococcus • **carbapenems ↓ valproate** • **metronidazole**: disulfiram-like, ↑ INR, neuropathy • tigecycline ↑ mortality.
32. **CDI risk:** clindamycin, FQs, cephalosporins, carbapenems, PPIs, age, hospitalization • **alcohol sanitizer doesn't kill spores → soap and water**, bleach, contact precautions.
33. **CDI testing:** only ≥ 3 unformed stools/24 h, no laxatives in 48 h • **no test of cure**, don't test asymptomatic.
34. **CDI severity:** non-severe WBC ≤ 15,000 and SCr < 1.5 • severe WBC > 15,000 or SCr > 1.5 • **fulminant** hypotension/shock, ileus, megacolon.

35. **CDI initial: fidaxomicin 200 mg BID × 10 d (preferred)** or **vancomycin 125 mg PO QID × 10 d** • metronidazole only if others unavailable (non-severe).
36. **Fulminant CDI:** vancomycin 500 mg PO/NG QID + metronidazole 500 mg IV q8h (+ rectal vancomycin if ileus) + surgery consult.
37. **Recurrence:** fidaxomicin or vancomycin taper/pulse • ≥ 2 recurrences → FMT, vancomycin → rifaximin • Vowst (oral) / Rebyota (rectal) after antibiotics • **IV vancomycin doesn't work** • avoid loperamide • stop PPIs • probiotics not recommended.

22. ID 3: Fungal Infections and Immunizations

Predicted for your exam (likelihood: HIGH). TB + aspergillosis 2022–23 and 2025; candidemia; vaccines in HTN, COPD and RA stations.

1. ★ Voriconazole + rifampin contraindicated → liposomal amphotericin B 3–5 mg/kg; amphotericin infusion reactions, saline loading, K/Mg.
2. ★ Candida krusei/glabrata → echinocandin; candidemia: remove line, eye exam, 14 days after first negative culture.
3. ★ Oral thrush: clotrimazole troche or nystatin; fluconazole for moderate–severe/esophageal.
4. ★ Adult vaccine catch-up: Tdap then Td q10y, influenza, PCV20/21, HBV 19–59, zoster ≥50, RSV, COVID-19.
5. ★ No live vaccines on biologics/high-dose steroids; give them ≥4 weeks before starting; infant BCG delayed after in-utero TNFi.
6. Azole interactions: QT, CYP3A4 (statins, tacrolimus, warfarin).

Exam Traps and Frequently Tested Points

1. **AmB** binds ergosterol • **azoles** inhibit 14-α-demethylase • **echinocandins** inhibit β-1,3-D-glucan synthase (cell wall) • **terbinafine** squalene epoxidase • **griseofulvin** microtubules • **flucytosine** → 5-FU.
2. **AmB:** infusion reactions (premedicate; meperidine for rigors), **nephrotoxicity** (saline loading), **↓ K, ↓ Mg** • conventional max **1.5 mg/kg** • D5W only • lipid forms less toxic.
3. **Fluconazole:** IV = PO, renal dose adjust, great urine/CSF penetration • **no C. krusei, no molds** • ↑ warfarin/phenytoin (2C9).
4. **Itraconazole:** capsule with food + acidic drink (no PPIs), solution empty stomach • **boxed HF** • DOC histoplasmosis/blastomycosis.
5. **Voriconazole: DOC aspergillosis** • visual disturbances, hallucinations, photosensitivity/skin cancer, periostitis • TDM (2C19) • IV vehicle avoid CrCl < 50 • **no Mucor activity** • CI sirolimus.
6. **Mucormycosis** (DKA, deferoxamine): **surgery + L-AmB** → isavuconazole/posaconazole • voriconazole and echinocandins inactive.
7. **Posaconazole**: prophylaxis in AML/MDS and GVHD • DR tablets with food • **isavuconazole shortens QT**.
8. **Echinocandins:** IV only, no renal adjust, few DDIs • **not for UTI, CNS, eye** • caspofungin 70 → 50 mg • C. parapsilosis ↑ MIC.
9. **Candidemia: echinocandin first** → step down to fluconazole after 5–7 days if stable/susceptible • **remove CVC • eye exam • 14 days after first negative culture**.
10. **C. krusei** fluconazole-resistant • **C. glabrata** reduced susceptibility • **C. auris** MDR → echinocandin + contact precautions.
11. **Candiduria:** asymptomatic → don't treat (except neutropenia, VLBW infants, urologic procedure) • symptomatic → **fluconazole 200 mg × 2 weeks**.
12. **Thrush:** clotrimazole troches, miconazole buccal, nystatin → fluconazole if moderate–severe • **esophageal → systemic fluconazole 14–21 days**.
13. **VVC: fluconazole 150 mg once** or topical azole • complicated → 150 mg q72h × 3 • recurrent → weekly × 6 months • **pregnancy → topical 7 days** • C. glabrata → boric acid.
14. **Tinea capitis and onychomycosis need oral therapy** • terbinafine 6 weeks (fingernails) / 12 weeks (toenails) • griseofulvin with fatty meal (inducer, teratogenic).
15. **Tinea versicolor** = Malassezia → selenium sulfide, ketoconazole • not oral terbinafine/griseofulvin.

16. **Cryptococcal meningitis:** L-AmB + flucytosine × 2 weeks → fluconazole 400–800 × 8 weeks → 200 mg ≥ 1 year • **serial LPs** for ICP • **delay ART 4–6 weeks**.
17. **PCP:** TMP-SMX 15–20 mg/kg/day × 21 days + **prednisone if PaO2 < 70 or A–a ≥ 35** • prophylaxis CD4 < 200: TMP-SMX DS daily (dapsone, atovaquone, pentamidine) • check G6PD for dapsone/primaquine.
18. **Live vaccines:** MMR, varicella, MMRV, rotavirus, LAIV, yellow fever, oral typhoid, BCG • **CI in pregnancy and severe immunosuppression** (CD4 < 200, pred ≥ 20 mg ≥ 14 days).
19. **Two live vaccines:** same day or ≥ 28 days apart • **TST:** same day as MMR or wait 4 weeks • IVIG/blood delays MMR/varicella.
20. **Only true contraindication for all vaccines:** anaphylaxis to prior dose/component • mild illness and antibiotics are not contraindications • **egg allergy → any flu vaccine**.
21. **Infant IM site:** vastus lateralis • adults: deltoid • **never gluteal** (HepB, rabies) • MMR/varicella SC • varicella frozen • observe 15 min (syncope).
22. **Rotavirus:** first dose by 14 weeks 6 days, last by 8 months • intussusception • **HPV:** 2 doses if started < 15 years, 3 doses if ≥ 15 or immunocompromised.
23. **HepB at birth** (+ HBIG within 12 h if mother HBsAg+) • **DTaP 5 doses** • **MMR/varicella 12–15 months + 4–6 years** • **Tdap 11–12** • **MenACWY 11–12 + 16**.
24. **Influenza:** everyone ≥ 6 months yearly • 2 doses for first-time 6 months–8 years • **≥ 65 → high-dose/adjuvanted/recombinant** • LAIV only 2–49 healthy non-pregnant.
25. **Tdap every pregnancy (27–36 weeks)** • Td/Tdap every 10 years • dirty wound: booster if ≥ 5 years; TIG if < 3 doses/unknown.
26. **Shingrix:** all ≥ 50 (≥ 19 immunocompromised) • 2 doses 2–6 months • non-live • give even after shingles/Zostavax • reactogenic.
27. **Pneumococcal (2024):** all ≥ 50 and 19–49 with risk → **PCV20 or PCV21 alone**, or **PCV15 → PPSV23 ≥ 1 year** (8 weeks if immunocompromised).
28. **Hepatitis B:** all adults 19–59 (≥ 60 with risk) • Heplisav-B 2 doses • dialysis high dose • anti-HBs ≥ 10 in HCWs.
29. **RSV:** single dose ≥ 75 and 50–74 at risk • **pregnancy Abrysvo 32–36 weeks** or infant nirsevimab.
30. **Pregnancy:** give inactivated flu, Tdap, RSV, COVID • **avoid MMR, varicella, LAIV, HPV**.
31. **Asplenia:** pneumococcal, MenACWY, MenB, Hib ≥ 14 days before splenectomy • **eculizumab/ravulizumab** → meningococcal vaccines (ACWY + B).
32. **Hajj/Umrah: MenACWY required** ≥ 10 days before arrival (conjugate valid 5 years) • yellow fever certificate from endemic countries.
33. **Rabies PEP:** wound wash → **HRIG 20 IU/kg into wound** + vaccine days **0, 3, 7, 14** (+ 28 immunocompromised) deltoid • previously vaccinated → days 0, 3 only, no HRIG.
34. **Other PEP:** HBV → HBIG + vaccine • varicella → vaccine ≤ 3–5 days or VariZIG ≤ 10 days • measles → MMR ≤ 72 h or IG ≤ 6 days • hep A → vaccine/IG ≤ 2 weeks.
35. **MMR → febrile seizures (MMRV higher), ITP, rash 7–12 days** • rotavirus → intussusception • mRNA → myocarditis (young males) • DTaP encephalopathy → no more pertussis vaccine.

23. Cardio 1: Ischemic Heart Disease

Predicted for your exam (likelihood: VERY HIGH). STEMI/ACS asked in 2015, 2018–19, 2022–23, 2024 and 2025.

1. ★ ER drugs with doses: aspirin 162–325 chewed, ticagrelor 180, UFH 70–100 units/kg, NTG SL, O2 only if <90%, morphine only if refractory.
2. ★ PCI vs lysis justification and time goals; major lytic risk = ICH.
3. ★ Discharge list (DAPT 12 months, atorvastatin 80, β-blocker, ACEi, MRA if indicated, SL NTG, cardiac rehab).
4. ★ Fix the home list: fluvastatin → atorvastatin 80; venlafaxine → sertraline; metformin hold 48 h after contrast.
5. ★ Contrast AKI prevention and AKI after PCI management.
6. NSTEMI with HTN emergency and pulmonary edema: IV NTG, avoid nitroprusside, IV furosemide.

Exam Traps and Frequently Tested Points

1. **HR is the main determinant of myocardial O2 demand** • β-blockers ↓ demand • nitrates ↓ preload • CCBs ↓ afterload/vasospasm.
2. **Stable angina** = exertional, relieved by rest/NTG • **unstable angina** = rest/new/crescendo, troponin negative • **NSTEMI** = troponin positive, no ST elevation • **STEMI** = ST elevation.
3. **Variant (Prinzmetal) angina → CCBs + nitrates**, avoid nonselective β-blockers.
4. **Hold caffeine/theophylline 12–24 h before adenosine/regadenoson stress testing**.
5. **CCD event reduction:** aspirin 75–100 mg (clopidogrel if intolerant), high-intensity statin, BP < 130/80, SGLT2i/GLP-1 RA in diabetes, ACEi/ARB if HTN/DM/CKD/EF ≤ 40%.
6. **β-blocker long term only if** LVEF ≤ 50%, MI within 1 year, or other indication (angina, arrhythmia, HTN).
7. **Antianginal first-line: β-blocker or CCB** → add long-acting nitrate or ranolazine • avoid β-blocker + verapamil/diltiazem • avoid IR nifedipine.
8. **β-blocker target HR 55–60** • no ISA • never stop abruptly • avoid in cocaine ACS and vasospastic angina.
9. **Nitrate tolerance → 10–14 h nitrate-free interval** (ISMN ER once in AM, patch off at night, ISDN TID with 14-h gap).
10. **Nitrates + PDE-5 inhibitors = contraindicated** (sildenafil/vardenafil 24 h, tadalafil 48 h) • also riociguat • RV infarction, SBP < 90.
11. **SL NTG 0.4 mg:** sit, repeat q5 min up to 3 doses, **call 911 if no improvement 5 min after the first dose** • original glass container.
12. **Ranolazine:** no effect on HR/BP • QT • **CI strong CYP3A4 inhibitors/inducers and cirrhosis** • simvastatin ≤ 20, metformin ≤ 1,000 • max 1,000 mg BID.
13. **ACS early therapy:** aspirin 162–325 chewed, NTG, O2 only if SpO2 < 90%, P2Y12, anticoagulant, high-intensity statin, oral β-blocker within 24 h, ACEi within 24 h (anterior MI/EF ≤ 40%).
14. **β-blocker contraindications in ACS:** HF, low output, cardiogenic shock risk, PR > 0.24 s, 2nd/3rd-degree block, active asthma • no IV β-blocker if shock risk.
15. **Avoid NSAIDs** after MI • morphine only for refractory pain (delays P2Y12 absorption).
16. **STEMI:** primary PCI if FMC-to-device ≤ 90 min (≤ 120 with transfer) • otherwise **fibrinolysis within 30 min** (symptoms < 12 h) → transfer, angiography 3–24 h.
17. **Fibrinolytics:** tenecteplase single weight-based bolus (half dose ≥ 75) • alteplase 15 mg + 0.75 mg/kg + 0.5 mg/kg • **never in NSTE-ACS** • know absolute contraindications.
18. **With fibrinolysis:** clopidogrel 300 mg (no load if > 75 → 75 mg) + enoxaparin (≥ 75: no bolus, 0.75 mg/kg).
19. **NSTE-ACS invasive timing:** immediate < 2 h (refractory angina, instability) • early < 24 h (GRACE > 140).
20. **Clopidogrel:** CYP2C19 prodrug, poor metabolizers ↓ effect, avoid omeprazole/esomeprazole (use pantoprazole) • 600 mg load for PCI.
21. **Prasugrel:** CI with prior stroke/TIA, avoid ≥ 75 years, 5 mg if < 60 kg, PCI only • **ticagrelor:** aspirin ≤ 100 mg, dyspnea, bradycardia, ↑ uric acid, CYP3A4.
22. **Hold before CABG:** clopidogrel 5 days, ticagrelor 3–5 days, prasugrel 7 days.
23. **DAPT 12 months after ACS** (shorten to 1–3 months if high bleeding risk; P2Y12 monotherapy after ≥ 1 month reasonable) • elective PCI in CCD: 6 months clopidogrel + aspirin.
24. **AF + PCI:** triple therapy ≤ 1 week (up to 30 days) → OAC (DOAC) + clopidogrel to 12 months → OAC alone • clopidogrel is the P2Y12 of choice.
25. **Anticoagulants:** UFH 60 units/kg + 12 units/kg/h • enoxaparin 1 mg/kg q12h • bivalirudin (PCI, HIT) • **fondaparinux not alone for PCI** • GP IIb/IIIa bailout only.
26. **Discharge after MI:** aspirin, P2Y12, high-intensity statin (LDL < 55–70; add ezetimibe → PCSK9i), β-blocker (EF ≤ 40%), ACEi/ARB, MRA (EF ≤ 40% + HF/DM), SL NTG, cardiac rehab.
27. **ICD** if LVEF ≤ 35% ≥ 40 days post-MI (≥ 90 days post-revascularization) on GDMT.
28. **Cocaine ACS:** benzodiazepines + nitrates/CCB, no β-blockers • **RV infarct:** fluids, avoid nitrates/diuretics.
29. **Ibuprofen blocks aspirin's antiplatelet effect** → aspirin 30 min before or 8 h after ibuprofen.

24. Cardio 2: Heart Failure, Arrhythmias and AF

Predicted for your exam (likelihood: VERY HIGH). AF then STEMI asked 2018–19, 2021, 2022–23, 2024; HFrEF 2020, 2024, 2025.

1. ★ CHA2DS2-VASc and HAS-BLED calculation (recalculate after MI).
2. ★ Stop flecainide (CAST) and diltiazem after MI; DOAC + clopidogrel; aspirin ≤1 week; OAC alone at 12 months.
3. ★ Apixaban dose criteria and warfarin → DOAC conversion INR thresholds.
4. ★ HFrEF: stage + NYHA, 4 pillars with start/target doses, ACEi → ARNI 36-h washout, stop amlodipine, spironolactone checks and monitoring.
5. ★ Amiodarone interactions (warfarin, digoxin, simvastatin) and monitoring; QT patient → doxycycline not azithromycin.
6. ADHF warm-wet: IV furosemide 1–2.5× home dose, diuretic resistance options; digoxin level 0.5–0.9.

Exam Traps and Frequently Tested Points

1. **HFrEF ≤ 40%**, HFmrEF 41–49%, **HFpEF ≥ 50%**, HFimpEF (continue GDMT) • **stages A–D never go backward**; NYHA I–IV can change.
2. **Sacubitril/valsartan ↑ BNP** → monitor with NT-proBNP • BNP ↑ in CKD, ↓ in obesity.
3. **Four pillars for HFrEF: ARNI (or ACEi/ARB) + β-blocker + MRA + SGLT2i** — start early, titrate to target • loop diuretics only for symptoms.
4. **ARNI:** 36-h washout from ACEi • CI angioedema, pregnancy • start 49/51 BID (24/26 if naive/eGFR < 30) → 97/103 BID.
5. **Only carvedilol, metoprolol succinate, bisoprolol** in HFrEF • start low go slow when euvolemic • carvedilol with food, target 25 BID (50 BID > 85 kg).
6. **MRA:** K < 5.0, eGFR > 30 • check K at 3 days, 1 week • spironolactone → gynecomastia (switch eplerenone) • eplerenone CYP3A4.
7. **SGLT2i (dapagliflozin/empagliflozin 10 mg)** for HFrEF, HFmrEF, HFpEF regardless of diabetes • eGFR ≥ 20–25.
8. **H-ISDN** for Black patients NYHA III–IV on GDMT or ACEi/ARB/ARNI intolerance • hydralazine lupus.
9. **Ivabradine:** sinus rhythm, HR ≥ 70 on max β-blocker, LVEF ≤ 35% • phosphenes, bradycardia • **vericiguat:** worsening HF, CI nitrates/PDE-5i.
10. **Digoxin:** ↓ hospitalizations, not mortality • level 0.5–0.9 • toxicity: N/V, yellow-green vision, arrhythmias • ↑ by hypoK/hypoMg • **amiodarone → ↓ digoxin 50%** • DigiFab.
11. **IV iron** if ferritin < 100 or 100–299 + TSAT < 20% • **ICD** LVEF ≤ 35% • **CRT** LVEF ≤ 35% + LBBB QRS ≥ 150.
12. **Avoid in HFrEF:** NSAIDs, diltiazem/verapamil, TZDs, saxagliptin/alogliptin, class Ic, dronedarone (NYHA IV/recent decompensation), itraconazole, cilostazol.
13. **Loop equivalence: furosemide 40 PO = 20 IV = torsemide 20 = bumetanide 1** • furosemide ~50% bioavailable • metolazone for resistance • ethacrynic acid for sulfa allergy.
14. **ADHF:** warm/wet → IV diuretics (1–2.5× oral dose IV) ± vasodilators • cold/wet → inotropes (dobutamine, milrinone — renal adjust, hypotension) • continue β-blocker unless shock.
15. **Nitroprusside → cyanide/thiocyanate toxicity** • HF education: daily weights (> 2–3 lb/day or 5 lb/week), sodium restriction, avoid NSAIDs.
16. **HFpEF:** SGLT2i, diuretics, BP control, MRA/finerenone, ARNI/ARB (may consider), GLP-1/tirzepatide in obesity.
17. **Class Ic (flecainide, propafenone) CI in structural heart disease/CAD (CAST)** • give with AV-nodal blocker • pill-in-pocket.
18. **Amiodarone:** pulmonary, thyroid, liver, eyes, skin (blue-gray), photosensitivity • TFT/LFT every 6 months • ↓ warfarin 30–50%, ↓ digoxin 50%, simvastatin ≤ 20.
19. **Dofetilide and sotalol:** start inpatient 3 days, CrCl dosing, torsades • dofetilide CI with verapamil, HCTZ, cimetidine, trimethoprim, ketoconazole, dolutegravir.
20. **Dronedarone:** CI permanent AF and NYHA IV/recent decompensated HF • procainamide → lupus, NAPA • disopyramide anticholinergic.

21. **Adenosine 6 → 12 mg rapid push** • ↓ dose with dipyridamole, carbamazepine, transplant, central line • theophylline/caffeine ↓ effect • bronchospasm.
22. **WPW with AF: avoid AV-nodal blockers** (adenosine, BB, CCB, digoxin, IV amiodarone) → procainamide/ibutilide/cardioversion.
23. **Torsades → Mg 2 g IV**, stop QT drugs, K > 4, Mg > 2, overdrive pacing/isoproterenol • QTc > 500 = high risk • know QT drug list.
24. **Symptomatic bradycardia:** atropine 1 mg (max 3 mg) → pacing/dopamine/epinephrine • drug causes: BB, CCB, digoxin, amiodarone, clonidine, donepezil.
25. **CHA2DS2-VASc:** age ≥ 75 and stroke = 2 points • **OAC if ≥ 2 men / ≥ 3 women** • consider 1 men / 2 women • **aspirin not recommended**.
26. **DOACs preferred** except mechanical valves and moderate–severe mitral stenosis (warfarin) • **HAS-BLED** modifies risk factors, doesn't exclude OAC • LAAO if OAC contraindicated.
27. **DOAC AF doses:** apixaban 5 BID (2.5 if 2 of 3: ≥ 80, ≤ 60 kg, SCr ≥ 1.5) • rivaroxaban 20 with evening meal (15 if CrCl 15–50) • dabigatran 150 BID (75 if CrCl 15–30) • edoxaban 60 (30 if CrCl 15–50; not if > 95).
28. **Cardioversion:** AF ≥ 48 h → 3 weeks OAC before (or TEE) + **4 weeks after for everyone** • unstable → immediate cardioversion.
29. **Rate control target < 100–110** • LVEF > 40%: BB or diltiazem/verapamil • **LVEF ≤ 40%: BB ± digoxin (no non-DHP CCB)** • IV diltiazem 0.25 mg/kg.
30. **Rhythm control:** ablation first-line in selected (HFrEF, paroxysmal) • no structural disease → flecainide/propafenone/dronedarone/sotalol • CAD → dronedarone/sotalol/dofetilide • HFrEF → amiodarone/dofetilide.
31. **No bridging for DOACs** • warfarin bridging only for mechanical valves/high risk • AF + PCI → DOAC + clopidogrel.
32. **AF risk-factor modification:** weight loss ≥ 10%, alcohol reduction, OSA treatment, exercise, BP control • check TSH.

25. Cardio 3: Hypertension and Dyslipidemia

Predicted for your exam (likelihood: VERY HIGH). Stage 2 HTN + prediabetes 2020 and 2024; dyslipidemia 2018–19, 2024.

1. ★ Stage 2 → two first-line agents; goal <130/80; lifestyle numbers (DASH, sodium, weight, exercise).
2. ★ Statin intensity by risk (14% → moderate; ASCVD/DM with risk factors → high); statin doses table.
3. ★ PCSK9 inhibitors names/doses and when NOT to add; aspirin not for primary prevention.
4. ★ Statin side effects and stop rules (CK >10× with symptoms, ALT >3× persistent); simvastatin caps with amiodarone/diltiazem.
5. HTN emergency targets and agents; DM + HTN → ACEi/ARB; ADA LDL <70.

Exam Traps and Frequently Tested Points

1. **BP categories:** normal < 120/80 • elevated 120–129/< 80 • **stage 1 130–139 or 80–89** • **stage 2 ≥ 140 or ≥ 90** • use the higher category.
2. **Measure correctly:** seated 5 min, back supported, feet flat, arm at heart level, correct cuff, average ≥ 2 readings on ≥ 2 occasions • confirm with home/ambulatory BP.
3. **White-coat HTN** (office ↑, home normal) → monitor • **masked HTN** (office normal, home ↑) → treat.
4. **Goal < 130/80 for all adults** (SBP < 120 encouraged if tolerated).
5. **Stage 1:** drugs if CVD, diabetes, CKD, or 10-year risk ≥ 7.5% (PREVENT; 10% ASCVD in 2017) • otherwise lifestyle 3–6 months • **stage 2: 2 drugs** (single-pill combination).
6. **Lifestyle:** DASH (~11 mmHg, largest), weight loss (1 mmHg/kg), Na < 1,500 or ↓ 1,000 mg, K 3,500–5,000 mg, exercise 90–150 min/week, alcohol ≤ 1–2 drinks.
7. **First-line: thiazide (chlorthalidone), ACEi, ARB, DHP CCB • Black patients (no CKD/HF): thiazide or CCB** • β-blockers only for compelling indications.
8. **Never ACEi + ARB (+ aliskiren)** • best SPC: ACEi/ARB + CCB.
9. **Pregnancy:** labetalol, nifedipine ER, methyldopa • **CI ACEi/ARB/ARNI/aliskiren** • aspirin 81 mg from 12 weeks for preeclampsia prevention.

10. **CKD with albuminuria → ACEi/ARB** (accept SCr ↑ ≤ 30%) • **gout → losartan**, avoid thiazides • **osteoporosis → thiazide • BPH → α-blocker (not monotherapy) • asthma → avoid nonselective β-blockers**.
11. **Thiazides:** ↓ K, ↓ Na, ↓ Mg, **↑ Ca, ↑ uric acid, ↑ glucose, ↑ lipids** • ineffective eGFR < 30 (except chlorthalidone/metolazone) • ↑ lithium.
12. **ACEi:** cough (bradykinin → switch to ARB), angioedema (never rechallenge), hyperK, ↑ SCr, CI pregnancy and bilateral RAS • check K/SCr in 2–4 weeks.
13. **DHP CCB:** ankle edema (↓ by adding ACEi/ARB), gingival hyperplasia, reflex tachycardia • **non-DHP:** bradycardia, constipation (verapamil), CI HFrEF, CYP3A4 inhibitors.
14. **β-blockers:** mask hypoglycemia (not sweating), bronchospasm (nonselective), **taper — no abrupt stop • clonidine rebound HTN** (taper; stop β-blocker first).
15. **α1-blockers:** first-dose syncope → bedtime • **hydralazine** lupus + reflex tachycardia (add β-blocker + diuretic) • **minoxidil** hirsutism, pericardial effusion • **methyldopa** hemolytic anemia (pregnancy).
16. **Resistant HTN:** ≥ 3 drugs incl. diuretic (or ≥ 4 controlled) • rule out non-adherence, white coat, drugs (NSAIDs, decongestants, OCs) • screen **primary aldosteronism**, OSA • **switch to chlorthalidone → add spironolactone**.
17. **Drug-induced HTN:** NSAIDs, decongestants, stimulants, OCs, steroids, cyclosporine/tacrolimus, EPO, venlafaxine, VEGF inhibitors, licorice, alcohol, cocaine.
18. **Severe HTN without organ damage (urgency):** oral therapy, lower over 24–48 h, **no IR nifedipine • emergency:** IV agents, ↓ ≤ 25% in 1st hour → 160/100–110 over 2–6 h → normal over 24–48 h.
19. **Aortic dissection: SBP < 120 within 20 min, β-blocker first (esmolol/labetalol)** • preeclampsia/pheochromocytoma: SBP < 140 in first hour • stroke: < 185/110 before tPA.
20. **Nitroprusside:** cyanide (hepatic) / thiocyanate (renal) toxicity, protect from light • **clevidipine:** CI soy/egg allergy • **phentolamine:** pheochromocytoma, cocaine, MAOI crisis.
21. **LDL = TC − HDL − TG/5** (invalid TG > 400) • non-HDL = TC − HDL • measure Lp(a) once • TG ≥ 500 → pancreatitis risk.
22. **Four statin groups:** ASCVD → high-intensity • **LDL ≥ 190 → high-intensity** • diabetes 40–75 → moderate (high if risk factors) • primary prevention by 10-year risk (7.5–19.9% moderate, ≥ 20% high).
23. **Risk enhancers:** family history, LDL ≥ 160, CKD, metabolic syndrome, preeclampsia/early menopause, RA/psoriasis/HIV, South Asian, TG ≥ 175, hs-CRP ≥ 2, Lp(a) ≥ 50, apoB ≥ 130, ABI < 0.9.
24. **CAC = 0 → may withhold statin** (except diabetes, smokers, family history) • CAC ≥ 100 → statin.
25. **High-intensity = atorvastatin 40–80, rosuvastatin 20–40** • moderate: atorva 10–20, rosuva 5–10, simva 20–40, prava 40–80, lova 40–80, pitava 1–4.
26. **Simvastatin/lovastatin in the evening** (lovastatin with food) • atorva/rosuva any time • lipid panel 4–12 weeks after start • no routine LFT/CK monitoring.
27. **SAMS:** hold → rechallenge lower dose/different statin/alternate-day • risk: high dose, elderly, hypothyroidism, CKD, DDIs • rhabdo CK > 10× ULN.
28. **Statin DDIs:** CYP3A4 (simva, lova, atorva) — avoid strong inhibitors, gemfibrozil, cyclosporine • **simvastatin ≤ 10 (verapamil, diltiazem), ≤ 20 (amiodarone, amlodipine, ranolazine)** • rosuvastatin 5 mg in Asians.
29. **Add-on order:** statin → **ezetimibe** → **PCSK9 inhibitor** (or inclisiran) • **bempedoic acid** for statin intolerance (gout, tendon rupture).
30. **Bile acid sequestrants:** ↑ TG (CI > 400), constipation, bind drugs (1 h before / 4–6 h after) • pregnancy-safe • colesevelam ↓ A1C.
31. **Fibrates for TG ≥ 500 • gemfibrozil + statin = avoid** (fenofibrate OK) • gallstones, ↑ SCr • niacin: flushing (aspirin 30 min before), hyperglycemia, gout — no added CV benefit.
32. **Icosapent ethyl 2 g BID** → ↓ MACE (TG 150–499 + ASCVD/DM on statin) • AF and bleeding risk.
33. **Statins in pregnancy → stop** • dialysis → don't start • HIV → pitavastatin • FH children from 8–10 years.

26. Therapeutic Drug Monitoring

Predicted for your exam (likelihood: VERY HIGH). Vancomycin and gentamicin TDM asked in ID and critical care 2018–2025; anti-Xa in pregnancy.

1. ★ Vancomycin: load 25 mg/kg, AUC 400–600, HD pre-dialysis level 15–20, post-HD level unreliable; action for level >20 or <10.
2. ★ Gentamicin: extended-interval 5–7 mg/kg + Hartford random level 6–14 h; synergy 1 mg/kg q8h trough <1; conventional peak/trough timing.
3. ★ Step-by-step calc: SCr µmol/L ÷ 88.4 → CrCl (CG) → IBW/AdjBW → dose.
4. ★ Enoxaparin anti-Xa peak timing and targets; UFH aPTT/anti-Xa 0.3–0.7.
5. Digoxin 0.5–0.9 ng/mL (amiodarone halves dose); valproate falls with carbapenems; INR as warfarin TDM.

Exam Traps and Frequently Tested Points

1. **TDM criteria:** narrow therapeutic index, high PK variability, concentration–effect relationship, no easy clinical endpoint, assay available • treat the patient, not the level.
2. **Steady state = 4–5 half-lives** (loading dose gets you there faster) • **trough = just before next dose** • **t½ = 0.693/ke** • **LD = Vd × C / (S × F)**.
3. **Aminoglycosides:** concentration-dependent • peak efficacy, trough toxicity • gent/tobra **peak 5–10, trough < 1–2** • amikacin **peak 20–35, trough < 4–8** • synergy peak 3–4.
4. **Peak = 30 min after end of 30-min infusion** • extended-interval → **random level 6–14 h + Hartford nomogram** • dose on IBW (AdjBW if obese).
5. **Extended-interval aminoglycoside contraindicated:** pregnancy, burns, ascites, CrCl < 20, endocarditis synergy.
6. **Vancomycin: AUC 400–600** (Bayesian or 2 levels) • trough-only (15–20) no longer recommended • load 20–35 mg/kg • AKI ↑ with pip-tazo • oral vanco → no TDM.
7. **Digoxin 0.5–0.9 ng/mL in HF** • sample ≥ 6–8 h post-dose • hypoK/hypoMg/hyperCa ↑ toxicity • amiodarone → ↓ digoxin 50% • levels unreliable after DigiFab • DLIS in renal failure.
8. **Phenytoin 10–20 total / 1–2 free** • **nonlinear** (small dose ↑ → big level ↑) • **albumin correction C/(0.2 × alb + 0.1)** (0.1 × alb + 0.1 in ESRD).
9. **Phenytoin toxicity:** nystagmus > 20 → ataxia > 30 → lethargy > 40 • IV ≤ 50 mg/min (hypotension, arrhythmias) • fosphenytoin ≤ 150 mg PE/min • phenytoin sodium = 92% (S 0.92) • hold tube feeds.
10. **Carbamazepine 4–12** • **autoinduction** → recheck at 3–4 weeks • hyponatremia, HLA-B*1502 • ↑ with macrolides, azoles, grapefruit.
11. **Valproate 50–100** (mania up to 125) • saturable protein binding → free level • thrombocytopenia, hyperammonemia • **carbapenems ↓ VPA**.
12. **Phenobarbital 15–40** (SS 2–3 weeks) • ethosuximide 40–100 • primidone 5–12 • lamotrigine/levetiracetam monitored in pregnancy.
13. **Lithium:** acute 0.8–1.2, maintenance 0.6–1.0, elderly 0.4–0.8 • **12 h post-dose** • 5 days after change • > 1.5 coarse tremor/GI, > 2 confusion/ataxia, > 2.5 seizures → dialysis.
14. **↑ Lithium:** thiazides, ACEi/ARB, NSAIDs, dehydration, low-Na diet, renal impairment • **↓ lithium:** caffeine, theophylline, pregnancy, high Na • monitor TSH, SCr, Ca.
15. **Clozapine ≥ 350 ng/mL** • **smoking cessation ↑ clozapine ~50%** (CYP1A2) • fluvoxamine/cipro ↑ • seizures dose-related • ANC monitoring.
16. **Nortriptyline 50–150 (therapeutic window)** • TCA toxicity → QRS > 100 ms → sodium bicarbonate • 2D6 inhibitors ↑ levels.
17. **Theophylline 5–15** (classic 10–20) • > 20 GI/tachycardia, > 30 seizures/arrhythmias • aminophylline = 80% theophylline • ↑ cipro, macrolides, cimetidine, fluvoxamine, HF, liver disease, quitting smoking • ↓ smoking, rifampin, phenytoin, carbamazepine.
18. **Tacrolimus 5–15, cyclosporine 100–400, sirolimus 5–15 ng/mL** • **whole blood trough** • CYP3A4/P-gp DDIs (azoles, macrolides, diltiazem/verapamil, PIs, grapefruit ↑; rifampin, carbamazepine, phenytoin, St John's wort ↓).
19. **Tacrolimus:** nephrotoxicity, tremor, hyperglycemia, hyperK, hypoMg • **cyclosporine:** + gingival hyperplasia, hirsutism, HTN, hyperlipidemia • **sirolimus:** hyperlipidemia, poor wound healing.

20. **Lidocaine 1.5–5** (CNS toxicity, seizures > 8; ↑ in HF/liver disease) • **procainamide 4–10 + NAPA 15–25** (NAPA renal; fast acetylators).
21. **Voriconazole trough 1–5.5** (CYP2C19; > 5.5 hallucinations/hepatotoxicity) • **posaconazole > 0.7 prophylaxis / > 1 treatment • flucytosine peak 30–80** (> 100 marrow).
22. **HDMTX:** levels 24/48/72 h (< 10 / < 1 / < 0.1 µM) → leucovorin until < 0.05–0.1 • hydration + urine pH ≥ 7 • avoid NSAIDs, PPIs, penicillins, TMP-SMX • glucarpidase.
23. **Anticoagulant monitoring:** warfarin INR 2–3 • UFH aPTT 1.5–2.5× or anti-Xa 0.3–0.7 • LMWH anti-Xa (peak 4 h, 0.5–1) only in renal/obesity/pregnancy • DOACs not routine.
24. **Acetaminophen nomogram:** level 4–24 h after acute ingestion, line 150 mcg/mL at 4 h → NAC • **salicylate** > 30 toxic; dialysis > 90–100.
25. **Free levels** for phenytoin and valproate in hypoalbuminemia, uremia, pregnancy, elderly, ICU.
26. **Smoking** induces CYP1A2 (theophylline, clozapine, olanzapine) — quitting ↑ levels • **never use proportional (linear) dose adjustment for phenytoin**.

27. Clinical Scores and DOACs

Predicted for your exam (likelihood: VERY HIGH). CHA2DS2-VASc/HAS-BLED in every AF station; Child-Pugh, CURB-65, qSOFA, KDIGO, NYHA in other systems.

1. ★ CHA2DS2-VASc (age ≥75 and stroke = 2) and HAS-BLED with assumptions for missing data.
2. ★ DOAC dosing by indication/renal function; switching rules; reversal agents.
3. ★ Child-Pugh A/B/C cut-offs; CURB-65 ≥3 severe; severe CAP major/minor criteria.
4. ★ qSOFA is a screening prompt only; KDIGO AKI staging; NYHA vs ACC stage; GOLD groups; Truelove-Witts.
5. Wells score → D-dimer vs imaging; MELD for transplant priority; CKD-EPI vs CG.

Exam Traps and Frequently Tested Points

1. **CHA2DS2-VASc:** age ≥ 75 and stroke/TIA = **2 points** each • OAC if **≥ 2 men / ≥ 3 women** • female sex alone not an indication.
2. **HAS-BLED ≥ 3 = high bleeding risk** → fix modifiable factors (BP, NSAIDs/aspirin, alcohol, labile INR) — **never a reason by itself to withhold OAC**.
3. **HAS-BLED letters:** HTN (SBP > 160), Abnormal renal/liver (1 each), Stroke, Bleeding, Labile INR, Elderly > 65, Drugs/alcohol (1 each) • max 9.
4. **TIMI UA/NSTEMI**: 7 items (age ≥ 65, ≥ 3 risk factors, known CAD, aspirin in 7 days, ≥ 2 anginal episodes/24 h, ST deviation, + biomarker) • ≥ 5 high.
5. **GRACE > 140 → early invasive < 24 h** • HEART 0–3 discharge • Killip IV = cardiogenic shock • DAPT score ≥ 2 → prolonged DAPT • PRECISE-DAPT ≥ 25 → short DAPT.
6. **Wells DVT ≤ 1 / PE ≤ 4 → D-dimer** • PERC only in low probability • sPESI 0 → outpatient • Padua ≥ 4 → prophylaxis • Caprini ≥ 5 high • Khorana ≥ 2 → apixaban/rivaroxaban.
7. **4T score:** 0–3 low (HIT unlikely) • 4–5 intermediate • 6–8 high • timing 5–10 days.
8. **Dabigatran:** direct thrombin inhibitor, prodrug, **80% renal**, dialyzable, dyspepsia, don't open capsules, original bottle, **idarucizumab 5 g**.
9. **Apixaban:** least renal (27%), least GI bleeding, OK in ESRD • AF 5 mg BID (2.5 if ≥ 2 of: ≥ 80 y, ≤ 60 kg, SCr ≥ 1.5) • VTE 10 mg BID × 7 d → 5 mg BID → 2.5 mg BID extended.
10. **Rivaroxaban:** 15/20 mg **with food** • AF 20 mg (15 if CrCl 15–50) • VTE 15 mg BID × 21 d → 20 mg → 10 mg extended • CAD/PAD 2.5 mg BID + aspirin.
11. **Edoxaban:** 60 mg (30 if CrCl 15–50, ≤ 60 kg, P-gp inhibitors) • **not in AF if CrCl > 95** • needs 5–10 days parenteral for VTE (like dabigatran).
12. **DOACs contraindicated** with mechanical valves, moderate–severe mitral stenosis (AF), triple-positive APS, pregnancy • avoid with rifampin, carbamazepine, phenytoin, St John's wort.
13. **Strong 3A4 + P-gp inhibitors:** avoid rivaroxaban; apixaban ↓ 50% • P-gp inhibitors ↑ dabigatran/edoxaban (dronedarone, ketoconazole, verapamil).
14. **Switching warfarin → DOAC:** start when INR < 2 (apixaban, dabigatran), < 3 (rivaroxaban), ≤ 2.5 (edoxaban) • LMWH → DOAC: 0–2 h before next LMWH dose.

15. **Periprocedural:** no bridging for DOACs • Xa inhibitors hold 1 day (low risk) / 2 days (high risk) • dabigatran longer if CrCl < 50 • resume 24 h / 48–72 h.
16. **Reversal:** dabigatran → idarucizumab (or dialysis) • apixaban/rivaroxaban → andexanet alfa or 4F-PCC 50 units/kg • vitamin K and protamine don't reverse DOACs.
17. **PT/aPTT unreliable for DOACs** → drug-calibrated anti-Xa / dilute thrombin time • DOAC CrCl by Cockcroft–Gault (actual body weight), recheck every CrCl/10 months if < 60.
18. **GCS ≤ 8 → intubate** • NIHSS 0–42 • ABCD2 ≥ 4 → DAPT 21 days after TIA • MMSE < 24 / MoCA < 26 abnormal.
19. **CURB-65:** 0–1 home, 2 admit, ≥ 3 ICU • qSOFA ≥ 2 (RR ≥ 22, AMS, SBP ≤ 100) • SOFA ↑ ≥ 2 = sepsis • Centor ≥ 2–3 → test.
20. **RASS goal −2 to 0** • BPS > 5 / CPOT > 2 = pain • ICDSC ≥ 4 delirium • CIWA ≥ 8–10 treat • COWS ≥ 8–12 start buprenorphine.
21. **Child-Pugh A 5–6, B 7–9, C 10–15** • MELD ≥ 15 transplant referral • West Haven grade 2 = asterixis • Maddrey ≥ 32 → prednisolone • Lille ≥ 0.45 stop steroids.
22. **Glasgow-Blatchford 0–1 → outpatient** • **Forrest Ia/Ib/IIa = high risk** → endoscopic therapy + IV PPI 72 h • LA grade C/D severe esophagitis.
23. **Ranson ≥ 3 / BISAP ≥ 3 = severe pancreatitis** • revised Atlanta severe = organ failure > 48 h • Truelove–Witts: ≥ 6 bloody stools + systemic sign → IV steroids.
24. **KDIGO AKI stage 1:** SCr ↑ ≥ 0.3 in 48 h or 1.5–1.9× • stage 3: ≥ 3× or ≥ 4.0 or RRT • FENa < 1% prerenal (FEUrea < 35% on diuretics) • Mentzer < 13 thalassemia.
25. **GOLD 1–4 by FEV1** (80/50/30) • groups A, B (LAMA+LABA), E (LAMA+LABA ± ICS if eos ≥ 300) • mMRC ≥ 2 / CAT ≥ 10 symptomatic • ACT ≥ 20 controlled.
26. **PHQ-9 ≥ 10 positive** (item 9 = suicide) • GAD-7 ≥ 10 • AUDIT ≥ 8 • CAGE ≥ 2 • Burch–Wartofsky ≥ 45 storm • DAS28 < 2.6 remission, > 5.1 high • FRAX ≥ 20% / ≥ 3%.

Note: predictions come from the patterns in your past-exam compilations; they show what is most likely, not a guarantee. Numbers follow current guidelines; if your course slides differ, use the course numbers.